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प्रहार Test Series

Test- 13

Physics + Chemistry

Solution

English

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BPSC TEST-13 (SOLUTIONS)

Q 1. Ans. C

Explanation:

- **Catenation** is the unique ability of carbon atoms to form covalent bonds with other carbon atoms to produce long chains, branched structures, rings, and complex networks. This property is the foundation of organic chemistry's extraordinary diversity, carbon can form single bonds (alkanes), double bonds (alkenes), and triple bonds (alkynes) with itself, creating millions of structurally distinct compounds. Silicon exhibits limited catenation, but no other element approaches carbon's ability in this regard. Examples include straight-chain compounds like n-hexane (C_6H_{14}), branched compounds like isobutane, and cyclic compounds like cyclohexane (C_6H_{12}) and benzene (C_6H_6).
- **Tetravalency** refers to carbon's ability to form four covalent bonds simultaneously with other atoms, including hydrogen, oxygen, nitrogen, halogens, and other carbons. While tetravalency enables catenation, it is a distinct concept describing valency rather than self-linkage. Example: methane (CH_4) demonstrates tetravalency with four C–H bonds.
- **Isomerism** is the phenomenon in which two or more compounds have the same molecular formula but different structural arrangements, resulting in different physical or chemical properties. Example: n-butane and isobutane both have the formula C_4H_{10} but differ in carbon chain arrangement.
- **Polymerisation** is the process by which small repeating units called monomers link together through covalent bonds to form large macromolecules called polymers. Example: ethene ($CH_2=CH_2$) monomers polymerise to form polyethylene (polythene), widely used in packaging.

Q 2. Ans. D

Explanation:

- **Statement 1 is incorrect:** The most abundant naturally occurring isotope of hydrogen is protium (^1H), containing one proton and no neutrons, constituting approximately 99.98% of naturally occurring hydrogen. Tritium (^3H) is radioactive, with a half-life of about 12.3 years, and occurs only in trace amounts.
- **Statement 2 is incorrect:** Hydrogen has a very high energy per unit mass (approximately 142 MJ/kg), significantly greater than conventional hydrocarbon fuels such as petrol, making it attractive as a fuel.
- **Statement 3 is incorrect:** Hydrogen can behave as both an oxidising agent and a reducing agent, depending on the reaction. For example, it acts as an oxidising agent when forming metal hydrides and as a reducing agent when reducing metal oxides.
- **Statement 4 is correct:** Deuterium (^2H or D) is the second isotope of hydrogen, containing one proton and one neutron in its nucleus, giving it a mass number of 2 while retaining hydrogen's atomic number of 1. Its chemical properties are essentially similar to ordinary hydrogen, though reaction rates may differ slightly due to the kinetic isotope effect. Deuterium occurs naturally in water as heavy water (D_2O) at a natural abundance of approximately 0.0156%, and is used as a moderator in certain nuclear reactors and in NMR spectroscopy.

Q 3. Ans. B

Explanation:

- In a plane mirror, the left side of the object appears as the right side in the image and vice versa. This effect is called lateral inversion.
- The mirror reverses the direction perpendicular to its surface while maintaining the vertical orientation unchanged. Hence, the image appears laterally inverted, though erect and virtual.

Q 4. Ans. D

Explanation:

- In longitudinal waves, compressions and rarefactions repeat periodically. The distance between two successive compressions or two successive rarefactions is called the wavelength.
- Amplitude is the maximum displacement of a particle of the medium from its mean or equilibrium position during wave motion.
- The time period is the time taken by a particle to complete one full oscillation or vibration.
- Frequency is the number of complete oscillations or vibrations made in one second.

Q 5. Ans. C

Explanation:

- **Pair 1 is correct:** Heat is defined as the transfer of thermal energy from one object or system to another due to a difference in temperature. The SI unit of heat is joules.

- **Pair 2 is correct:** Specific heat capacity is the amount of heat energy required to raise the temperature of a unit mass of a substance by one degree Celsius or one Kelvin. It measures a material's ability to store thermal energy. Its SI unit is joule per kilogram per kelvin.
- **Pair 3 is correct:** Temperature is a physical quantity that measures the degree of hotness or coldness of a substance. The SI unit of temperature is kelvin.
- **Pair 4 is incorrect:** Thermal conductivity is a material's intrinsic ability to transfer or conduct heat. It determines how quickly heat flows through a substance from a hotter area to a colder area. It is measured in **Watts per metre Kelvin,** not Newton.

Q 6. Ans. C

Explanation:

- Smelting is the pyrometallurgical process of extracting a metal from its ore by heating it with a suitable reducing agent, typically coke (carbon) or carbon monoxide, at high temperatures, causing the metal oxide to be chemically reduced to the free metal in molten form. The molten metal is then separated from the slag (waste material) and collected. Classic examples include the extraction of iron from haematite (Fe_2O_3) in a blast furnace using coke, and the extraction of copper from copper pyrites and zinc from zinc oxide using carbon as the reducing agent.
- Calcination is the process of heating an ore in the absence or limited supply of air below its melting point to remove volatile impurities such as moisture, carbon dioxide, or organic matter. It does not involve a reducing agent and does not yield the metal directly. Examples include heating limestone ($CaCO_3$) to produce quicklime (CaO) by expelling CO_2, and heating zinc carbonate ($ZnCO_3$) to obtain zinc oxide (ZnO) before further reduction.
- Electrorefining is a purification process, not an extraction process, in which an impure metal anode is dissolved electrolytically and pure metal is deposited at the cathode in an electrolytic cell. It is used to obtain high-purity metals after initial extraction. Examples include the electrorefining of copper, silver, gold, and nickel to achieve purities exceeding 99.9%.
- Concentration refers to the preliminary step of removing gangue (unwanted earthy material) from the ore before actual extraction begins, using physical or chemical methods. It does not involve chemical reduction or heating with reducing agents. Examples include froth flotation for sulphide ores like copper pyrites, magnetic separation for magnetite (Fe_3O_4), and gravity separation (hydraulic washing) for oxide ores.

Q 7. Ans. A

Explanation:

- **An isothermal process** is a thermodynamic process in which the temperature of the system remains constant throughout the process.
- **Isobaric Process** is a process carried out at constant pressure, while volume and temperature may change.
- **Isochoric Process** (also called isovolumetric process) occurs at constant volume. Since volume does not change, no mechanical work is done by the gas.
- **An adiabatic process** is a thermodynamic process in which no heat enters or leaves the system.

Q 8. Ans. B

Explanation:

- **Statement 1 is correct:** Regelation is the phenomenon in which ice melts under pressure and refreezes when the pressure is removed. In the wire experiment, pressure exerted by the wire lowers the melting point of ice, causing local melting. After the wire passes, water freezes again.
- **Statement 2 is correct:** Skating is possible because pressure exerted by the skates forms a thin layer of water beneath them. This water acts as a lubricant and reduces friction.
- **Statement 3 is incorrect:** In sublimation, a substance changes directly from the solid state to the vapour state without passing through the liquid state.
- **Statement 4 is correct:** Dry ice (solid CO_2) and iodine are common substances that undergo sublimation.

Q 9. Ans. C

Explanation:

- Avogadro's Law (1811) states that equal volumes of all gases, measured at the same temperature and pressure, contain equal numbers of molecules, regardless of the chemical nature, mass, or complexity of the gas. This was a foundational insight because it established that the number of gaseous particles, not their identity or mass, determines the volume occupied under identical conditions.

- The standard quantitative expression of this is the concept of molar volume; one mole of any gas at STP (0°C, 1 atm) occupies 22.4 litres, containing 6.022×10^{23} molecules (Avogadro's number), whether the gas is hydrogen, oxygen, carbon dioxide, or any other.

Q 10. Ans. D

Explanation:

- The correct order of increasing thermal conductivity is:
  - ✓ Bismuth (7.97 W/m·K) → Aluminium (237 W/m·K) → Copper (401 W/m·K) → Silver (429 W/m·K)
- **Bismuth** has one of the lowest thermal conductivities among pure elemental metals, making it the poorest thermal conductor among the four. This is due to its complex rhombohedral crystal structure and low free electron mobility.
- **Aluminium**, despite being a lightweight metal, is a reasonably good thermal conductor, a property exploited extensively in heat sinks, cookware, and aerospace applications.
- **Copper** is an excellent thermal conductor, second only to silver among pure metals, which is why it is the preferred material for heat exchangers, electrical wiring, and cooking vessels requiring rapid heat distribution.
- **Silver** has the highest thermal conductivity of any pure metal, making it the best thermal conductor among the four. However, its high cost limits its practical applications to specialised electronics and scientific instruments.

Q 11. Ans. B

Explanation:

- Platinum is commonly referred to as the "white noble metal", a designation that reflects both its characteristic silvery-white lustrous appearance and its exceptional chemical inertness. The term "noble metal" denotes metals that resist oxidation, corrosion, and chemical attack under ordinary conditions, and platinum exemplifies this property, remaining unaffected by air, moisture, and most acids, dissolving only in aqua regia (a mixture of concentrated nitric and hydrochloric acids).
- Platinum's nobility and rarity make it highly valuable, often commanding premium market prices. Its applications reflect this unique combination of properties; it is used in automobile catalytic converters to reduce exhaust emissions, as laboratory crucibles due to its high-temperature chemical inertness, in jewellery, in anticancer drugs such as cisplatin, and historically in the international kilogram prototype standard.

Q 12. Ans. D

Explanation:

- The displacement of one metal from its salt solution by another is governed by the reactivity series (activity series) of metals. A metal can displace another from its salt solution only if it is more reactive, meaning it has a greater tendency to lose electrons and form positive ions. The more reactive metal displaces the less reactive one by reducing the metal ions in solution back to their elemental form.
- For example, when iron is placed in copper sulphate solution:
  - ✓ $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- Iron displaces copper because iron is higher in the reactivity series. Similarly, zinc displaces copper, and copper displaces silver. This principle also underlies electrochemical cells, where the more reactive metal acts as the anode.

Q 13. Ans. C

Explanation:

- Rusting is an electrochemical oxidation process in which iron reacts with atmospheric oxygen in the presence of moisture to form hydrated iron(III) oxide ($Fe_2O_3 \cdot xH_2O$), commonly known as rust. The overall reaction can be represented as:
  - ✓ $4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$
- Since oxygen and water molecules from the surroundings chemically combine with iron, the product (rust) incorporates the mass of both iron and oxygen. By the Law of Conservation of Mass, the total mass of the rusted product is equal to the mass of iron plus the mass of oxygen that has been incorporated. The weight of the iron object therefore, increases as rusting progresses.
- This is consistent with the broader principle that any oxidation reaction, whether burning, rusting, or tarnishing, results in mass gain when the oxidised product remains on the surface, because atoms from the surrounding atmosphere are added to the material.

Q 14. Ans. B

Explanation:

- Galvanisation is the process of coating iron or steel with a thin layer of zinc to protect it from rusting.

Zinc protects iron through two complementary mechanisms operating simultaneously.

- The first is a simple physical barrier; the zinc coating prevents direct contact of the iron surface with moisture and oxygen, thereby inhibiting the electrochemical process responsible for rust formation.
- The second, and more important, mechanism is sacrificial protection (cathodic protection). Zinc is higher than iron in the reactivity series and therefore oxidises more readily. If the coating is scratched and both metals are exposed to moisture, zinc preferentially undergoes oxidation, sacrificing itself while protecting the iron from corrosion.
- The oxidised zinc forms a stable protective layer of zinc oxide (ZnO) and zinc carbonate ($ZnCO_3$), which further reduces corrosion. Because of this sacrificial action, even damaged galvanised surfaces continue to protect the underlying iron, making galvanisation particularly effective.

Q 15. Ans. A

Explanation:

- **Statement 1 is correct:** In diamond, each carbon atom is sp^3 hybridised and forms four strong covalent bonds with four neighbouring carbon atoms in a three-dimensional tetrahedral network. This rigid, interlocking structure extending in all directions makes diamond the hardest naturally occurring substance. In graphite, each carbon atom is sp^2 hybridised and forms three covalent bonds with three neighbouring carbon atoms arranged in flat hexagonal layers held together by weak van der Waals forces. These layers slide easily over one another, making graphite soft and useful as a lubricant.
- **Statement 2 is incorrect:** Graphite conducts electricity because each sp^2 hybridised carbon atom has one delocalised electron available for conduction. Diamond does not conduct electricity under normal conditions because all four valence electrons are involved in localised covalent bonding.
- **Statement 3 is incorrect:** It is diamond, not graphite, that possesses the stronger three-dimensional covalent network. Graphite's layers are held together only by weak intermolecular forces.
- **Statement 4 is incorrect:** Each carbon atom in diamond forms four covalent bonds, whereas each carbon atom in graphite forms three, making diamond the more rigidly bonded structure.

Q 16. Ans. D

Explanation:

- Strontium compounds, primarily strontium carbonate ($SrCO_3$) and strontium nitrate [$Sr(NO_3)_2$], are responsible for the characteristic bright red colour in fireworks. When strontium salts are heated in a pyrotechnic flame, strontium atoms absorb thermal energy and their electrons are excited to higher energy levels. As the electrons return to lower energy states, they emit photons of specific wavelengths corresponding to red light (approximately 600–700 nm). This is an application of atomic emission spectroscopy on a visible scale.
- The colours produced in fireworks and their responsible elements are as follows:
  - ✓ Red: Strontium compounds ($SrCO_3$, $Sr(NO_3)_2$)
  - ✓ Crimson/deep red: Lithium compounds (Li_2CO_3, LiCl)
  - ✓ Orange: Calcium compounds ($CaCl_2$, $CaCO_3$)
  - ✓ Yellow/golden: Sodium compounds ($NaNO_3$, NaCl), sodium's yellow emission is extremely intense and can dominate other colours even in trace amounts
  - ✓ Green: Barium compounds ($Ba(NO_3)_2$, $BaCl_2$)
  - ✓ Blue: Copper compounds ($CuCl_2$, CuO), one of the most difficult colours to produce reliably in pyrotechnics
  - ✓ White/silver: Magnesium or aluminium powder burning at high temperatures
  - ✓ Purple/violet: Typically produced by combining strontium (red) and copper (blue) compounds simultaneously.

Q 17. Ans. B

Explanation:

- Critical temperature (Tc) is defined as the temperature above which a gas cannot be liquefied by pressure alone, regardless of how much pressure is applied. At or below the critical temperature, intermolecular attractive forces become sufficiently significant relative to molecular kinetic energy that applying adequate pressure can bring molecules close enough together to produce liquefaction. The critical pressure (Pc) is the minimum pressure required to liquefy a gas exactly at its critical temperature.
- For example, CO_2 has a critical temperature of 31.1°C, above this temperature it cannot be liquefied regardless of pressure. Below this temperature, sufficient pressure can liquefy it successfully.

- Gases with relatively high critical temperatures, such as SO_2 (157.8°C) and NH_3 (132.4°C), can be liquefied comparatively easily at room temperature, whereas gases such as N_2 (−147°C) and O_2 (−118°C) must first be cooled below their critical temperatures before pressure-induced liquefaction becomes possible.

Q 18. Ans. C

Explanation:

- **Pair 1 is incorrectly matched:** Brass is an alloy of copper and zinc (not tin), typically containing 60–70% copper and 30–40% zinc. It is valued for its corrosion resistance, acoustic properties, and machinability, and is widely used in musical instruments, plumbing fittings, and decorative items.
- **Pair 2 is incorrectly matched:** Bronze is an alloy of copper and tin (not zinc), typically containing about 80–90% copper and 10–20% tin. It is harder than brass and has excellent wear resistance, historically significant as the defining metal of the Bronze Age. It is used in sculptures, bearings, and marine fittings.
- **Pair 3 is correctly matched:** Solder is correctly matched, it is an alloy of lead (Pb) and tin (Sn), typically in a ratio of approximately 1:2 (33% lead, 67% tin) for common soft solder. Its relatively low melting point (~183°C) makes it ideal for joining electronic components and plumbing connections without damaging the materials being joined. Lead-free solders using tin-silver-copper alloys have increasingly replaced traditional lead-tin solder in electronics due to environmental and health regulations.
- **Pair 4 is incorrectly matched:** Steel is an alloy of iron and carbon (not aluminium), with carbon content ranging from approximately 0.02% to 2.1% by weight. Different grades of steel, such as mild steel, stainless steel, and high-carbon steel, are produced by varying carbon content and adding alloying elements like chromium, nickel, and manganese.

Q 19. Ans. C

Explanation:

- Light waves are electromagnetic waves and are transverse in nature because their oscillations are perpendicular to the direction of propagation.
- Sound waves in air, compression waves in a spring, and ultrasonic waves in water are longitudinal waves because the particles oscillate parallel to the direction of propagation.

Q 20. Ans. A

Explanation:

- **Statement 1 is correct:** Faraday's First Law states that whenever there is a change in magnetic flux linked with a coil, an induced emf is produced in the coil. If the circuit is closed, this emf produces an induced current.
- **Statement 2 is incorrect:** Faraday's Second Law clearly states that the magnitude of induced emf is directly proportional to the rate of change of magnetic flux. A faster change in flux produces a larger emf, while a slower change produces a smaller emf. Therefore, only Statement 1 is correct.

Q 21. Ans. A

Explanation:

- The Faraday constant represents the total electric charge carried by one mole of electrons.
- Its standard value is approximately 9.65×10^4 coulomb per mole ($C\ mol^{-1}$).
- It is a fundamental constant used in electrochemistry, especially in electrolysis calculations.
- Faraday's constant is obtained by multiplying Avogadro's number (6.022×10 23 mol −1) with the charge of one electron (1.6×10−19 C). Hence,
  ✓ $F = N_A \times e$
- This shows that it is the total charge of all electrons present in one mole.

Q 22. Ans. D

Explanation:

- **Statement 1 is correct:** When a magnet is moved towards or away from a coil, the magnetic flux changes and induces current in the coil.
- **Statement 2 is correct:** According to Faraday's Law, an induced current is produced only when there is a change in magnetic flux linked with the circuit.
- **Statement 3 is correct:** Electromagnetic induction is the working principle of electric generators, where mechanical energy is converted into electrical energy.

Q 23. Ans. C

Explanation:

- **Statement 1 is correct:** Electrical resistance is the property of a material that opposes the flow of electric current through it. Greater resistance means lesser current flow through the conductor.
- **Statement 2 is correct:** The SI unit of resistance is ohm (Ω), represented by the Greek letter omega.

Resistance is measured using an instrument called an ohmmeter. Metals such as copper, silver and aluminium have low resistance and are therefore good conductors of electricity, whereas rubber, glass and plastic have very high resistance and act as insulators.

Q 24. Ans. A

Explanation:

- **Statement 1 is correct:** Capacitance is the property of a circuit or device to store electric charge. It is represented by the symbol C.

$$C = \frac{q}{V}$$

- **Statement 2 is correct:** A capacitor is made up of two conducting plates separated by an insulating material known as a dielectric material.
- **Statement 3 is correct:** Inductance is the property of an electric circuit that opposes any change in current flowing through the circuit. It is represented by the symbol LLL.
- **Statement 4 is incorrect:** The SI unit of inductance is henry (H), not ohm (Ω). Ohm is the SI unit of electrical resistance.

Q 25. Ans. D

Explanation:

- **Statement 1 is correct:** The inductance of a coil is influenced by factors such as the number of turns in the coil, its diameter, length and the type of core material used. A greater number of turns generally increases inductance.
- **Statement 2 is correct:** In an alternating current (AC) circuit, current changes continuously with time. Inductance opposes this change in current by producing a self-induced emf.
- **Statement 3 is correct:** Electric current flowing through a conductor creates a magnetic field around it. Whenever the current changes, the magnetic field strength also changes correspondingly. This changing magnetic field induces a voltage in the conductor, which opposes the change in current flow.

Q 26. Ans. C

Explanation:

- **Pair 1 is correct:** A battery is a combination of two or more electric cells connected to provide a higher voltage and energy.
- **Pair 2 is correct:** A galvanometer is a sensitive instrument used to detect and indicate the presence of electric current in a circuit.
- **Pair 3 is incorrect:** A resistor does not generate a magnetic field; it only opposes or limits the flow of electric current in a circuit. Magnetic fields are associated with current-carrying conductors, not resistors, as a function.
- **Pair 4 is correct:** A heating element is made of high-resistance material like nichrome. When current flows through it, electrical energy is converted into heat energy due to the heating effect of current.

Q 27. Ans. B

Explanation:

- A potentiometer has three terminals. When only two terminals are used (one fixed end and the wiper), the device functions by varying the length of the resistive track in the circuit, thereby changing the effective resistance.
- The wiper slides along the resistive track, increasing or decreasing the length of resistance wire included in the circuit, which directly changes the resistance value.
- When all three terminals are used, it acts as a voltage divider providing variable voltage. But in two terminal configuration, only resistance is being varied.
- Hence, in two terminal use, a potentiometer allows adjustment of resistance, not voltage, current, or power directly.

Q 28. Ans. B

Explanation:

- **Pipe earthing** is considered the most common, economical and effective method of earthing. In this method, a hollow GI pipe (generally about 38 mm in diameter and 2.5 m long) is buried vertically in the ground. The pit is filled with layers of charcoal and salt to improve the conductivity of the soil.
- Pipe earthing provides a low-resistance path for fault current to flow safely into the earth, thereby protecting electrical appliances and users from electric shock.
- **Plate earthing:** Involves burying a copper or GI plate deep in the ground, but is more expensive and less commonly used compared to pipe earthing.
- **Rod earthing:** Uses a solid metal rod driven into the earth; it is used in rocky or hard soil areas.
- **Strip earthing:** Uses metal strips laid horizontally in trenches and is used in specific industrial installations.

Q 29. Ans. C

Explanation:

- Iron is a **ferromagnetic material**, strongly attracted to magnetic fields and can be permanently magnetised.
- Aluminium is **paramagnetic**, showing weak attraction due to unpaired electrons.
- Copper is **diamagnetic**, meaning it is weakly repelled by magnetic fields.
- Lead is also **diamagnetic**, so pairing it with ferromagnetic is incorrect.

Q 30. Ans. A

Explanation:

- Statement 1 is correct: Paramagnetic materials (e.g., aluminium) are weakly attracted by a magnetic field due to unpaired electrons and do not retain magnetism once the field is removed.
- Statement 2 is incorrect: Diamagnetic materials (e.g., copper, lead) are weakly repelled by a magnetic field, not strongly attracted.
- Statement 3 is correct: Ferromagnetic materials (e.g., iron, cobalt, nickel) are strongly attracted by magnetic fields and can retain magnetism even after the external field is removed.

Q 31. Ans. C

Explanation:

- Fleming's Right-Hand Rule is used to determine the direction of induced current when a conductor moves in a magnetic field, which is the basic principle of electric generators.
- According to this rule, the three fingers of the right hand are held mutually perpendicular to each other:
  - ✓ The thumb represents the direction of motion of the conductor
  - ✓ The forefinger (Index finger) represents the direction of the magnetic field
  - ✓ The middle finger represents the direction of the induced current

Q 32. Ans. A

Explanation:

- **Statement 1 is correct:** Adding common salt (NaCl) to pure water causes boiling point elevation, a colligative property. Dissolved NaCl dissociates into Na^+ and Cl^- ions, increasing the number of solute particles in solution. This lowers the vapour pressure of water, requiring a higher temperature for the vapour pressure to equal atmospheric pressure, thereby increasing the boiling point.
- **Statement 2 is correct:** Adding methyl alcohol (methanol) to pure water causes freezing point depression, another colligative property. The dissolved methanol molecules interfere with the formation of the ice lattice, causing the solution to freeze at a temperature below 0°C.
- **Statement 3 is incorrect:** The addition of impurities to pure water generally lowers the freezing point, not raises it. This is the phenomenon of freezing point depression, a standard colligative property.

Q 33. Ans. C

Explanation:

- Oxides that react with both acids and bases are called amphoteric oxides. Aluminium oxide (Al_2O_3) and Zinc oxide (ZnO) are the classic examples of amphoteric oxides.
- Al_2O_3 reacts with acids and bases as follows:
  - ✓ With acid: $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
  - ✓ With base: $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$
- ZnO similarly behaves as both acidic and basic:
  - ✓ With acid: $ZnO + H_2SO_4 \rightarrow ZnSO_4 + H_2O$
  - ✓ With base: $ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$

Q 34. Ans. B

Explanation:

- **Pair 2 is incorrectly matched:** The acid predominantly found in citrus fruits is citric acid, not malic acid. Malic acid is primarily associated with apples, cherries, and plums, it is responsible for the sharp tartness characteristic of unripe apples.

Q 35. Ans. C

Explanation:

- **Pair 1 is incorrectly matched:** Baking soda is $NaHCO_3$ (sodium hydrogen carbonate / sodium bicarbonate), not $Na_2CO_3{\cdot}10H_2O$. The given formula belongs to washing soda.
- **Pair 2 is incorrectly matched:** Washing soda is $Na_2CO_3{\cdot}10H_2O$ (sodium carbonate decahydrate), not $NaHCO_3$. The given formula corresponds to baking soda.
- **Pair 3 is correctly matched:** Bleaching powder is represented by $CaOCl_2$ (calcium oxychloride), produced by passing chlorine over dry slaked lime [$Ca(OH)_2$]. Its bleaching and disinfecting action arises from the release of chlorine in the presence of moisture.
- **Pair 4 is incorrectly matched:** Plaster of Paris is $CaSO_4{\cdot}½H_2O$ (calcium sulphate hemihydrate), not $CaCO_3$. $CaCO_3$ is calcium carbonate, found in limestone, marble, and chalk.

Q 36. Ans. B

Explanation:

- **Sublimation** is the most suitable method for separating sand and naphthalene because naphthalene is a sublime substance; it directly converts from solid to vapour upon heating without passing through a liquid phase, and re-solidifies upon cooling. Since sand does not sublime, heating the mixture causes naphthalene vapours to rise and collect on a cooled surface above, leaving pure sand behind. This property makes sublimation uniquely effective for separating sublime substances from non-sublime ones. Other common examples of sublime substances include camphor, iodine, ammonium chloride, and dry ice (CO_2).
- **Filtration** is a technique used to separate an insoluble solid from a liquid by passing the mixture through a porous medium such as filter paper, which retains the solid while allowing the liquid to pass through. Common examples include separating chalk powder from water or tea leaves from tea.
- **Distillation** is used to separate miscible liquids with different boiling points by heating the mixture until the more volatile component vaporises, then condensing and collecting the vapour separately. A classic example is separating acetone from water or purifying alcohol.
- **Crystallisation** is used to purify soluble solids from solution by dissolving the impure substance in a hot solvent, then allowing slow cooling to form pure crystals of the desired compound while impurities remain in solution. It is used to purify substances like copper sulphate, sugar, and common salt.

Q 37. Ans. A

Explanation:

- Since both acids have the same molar concentration but different pH values, the difference in pH directly reflects their different degrees of dissociation. Acid A with pH 2 has $[H^+] = 10^{-2}$ mol/L, while Acid B with pH 4 has $[H^+] = 10^{-4}$ mol/L, meaning Acid A produces 100 times more hydrogen ions than Acid B at identical concentrations. Greater dissociation into H^+ ions at the same concentration is the defining characteristic of a stronger acid.

Q 38. Ans. A

Explanation:

- **Statement 1 is correct:** The resistance of a conductor increases with an increase in its length. Mathematically,
  - ✓ $R \propto l$
- **Statement 2 is correct:** Resistance decreases when the area of cross-section of the conductor increases. Thus,
  - ✓ $R \propto \frac{1}{A}$

Combining both relations,

✓ $$R = \rho \frac{l}{A}$$

- **Statement 3 is correct:** Resistivity is a characteristic property of a material and depends on the nature of the substance.
- **Statement 4 is incorrect:** Insulators such as rubber and glass have very high resistivity compared to metals. Metals and alloys possess very low resistivity and are therefore good conductors of electricity.

Q 39. Ans. B

Explanation:

- A semiconductor is a material whose electrical conductivity lies between that of a conductor and a non-conductor.
- Its conductivity can increase due to the addition of impurities or due to the temperature effect.
- The process of adding controlled impurities is called doping, and doped semiconductors are known as extrinsic semiconductors.
- Their resistance decreases as temperature increases, which is opposite to the behaviour of metals. Examples of semiconductors include Germanium, Selenium, Silicon, and Carbon.

Q 40. Ans. D

Explanation:

- Statement 1 is correct: In a purely resistive circuit, the energy supplied by the source is entirely dissipated in the form of heat. This phenomenon is known as the heating effect of electric current.
- Statement 2 is correct: The heating effect of electric current is practically used in appliances such as electric heaters, electric irons, toasters, kettles, and geysers.
- Statement 3 is correct: Electrical power supplied to a circuit is equal to the product of potential difference and current.
  - ✓ P=VI
- The heat produced in time t is given by H=VIt

Q 41. Ans. C

Explanation:

- **Statement 1 is correct:** Kirchhoff's rules are applied for analysing complicated electric circuits where simple series and parallel combination formulae are insufficient to determine currents and potential differences.
- **Statement 2 is correct:** While solving circuits, the direction of current is initially assumed. If the calculated current comes out positive, the current flows in the assumed direction. If it is negative, the actual current flows opposite to the assumed direction. Kirchhoff's rules are based on the principles of conservation of charge and conservation of energy in an electric circuit.

Q 42. Ans. A

Explanation:

- **Statement 1 is correct:** Metals are good conductors of electricity because they have very low resistivity, generally in the range of 10–8 Ωm to 10 –6 Ωm.
- **Statement 2 is correct:** Insulators such as glass and rubber possess extremely high resistivity, making them poor conductors of electricity.
- **Statement 3 is correct:** Semiconductors like silicon (Si) and germanium (Ge) have resistivity values intermediate between conductors and insulators.
- **Statement 4 is incorrect:** In most substances, electric current is carried by electrons, but in ionic crystals and electrolytic solutions, both positive and negative ions act as charge carriers.

Q 43. Ans. A

Explanation:

- Pair 1 is correct: The SI unit of electric current is ampere (A).
- Pair 2 is incorrect: The SI unit of resistance is ohm (Ω), not volt.
- Pair 3 is correct: The SI unit of resistivity is ohm metre (Ω m).
- Pair 4 is incorrect: The SI unit of electric potential difference is volt (V), whereas coulomb (C) is the SI unit of electric charge.

Q 44. Ans. B

Explanation:

- A dry cell is an electrochemical device that produces electricity through chemical reactions taking place inside the cell. Therefore, it converts chemical energy into electrical energy.
- In a dry cell, the chemical reaction between the electrolyte and electrodes creates a potential difference between the two terminals. When the cell is connected in a circuit, this potential difference causes the flow of electric current.
- Dry cells are portable, compact and widely used in devices such as torches, remotes, radios, clocks and toys.
- Option (A) is incorrect: Conversion of electrical energy into chemical energy occurs in rechargeable batteries during charging.
- Option (C) is incorrect: Mechanical energy into electrical energy conversion takes place in generators or dynamos.
- Option (D) is incorrect: Heat energy is not converted into electrical energy in a dry cell. Devices such as thermoelectric generators are examples where heat energy is converted into electrical energy.

Q 45. Ans. C

Explanation:

- **Statement 1 is correct:** A voltaic cell, also known as an electric cell, converts chemical energy into electrical energy by means of spontaneous chemical reactions occurring inside the cell. The chemical action creates a potential difference between the terminals, causing an electric current to flow in a connected circuit.
- **Statement 2 is correct:** The voltaic cell is named after Alessandro Volta, who invented the first electric battery known as the voltaic pile. The SI unit of potential difference, volt (V), is also named in his honour. Volta's invention marked a major development in the field of electricity because it provided a continuous source of electric current.

Q 46. Ans. C

Explanation:

- **Statement 1 is correct:** Dalton's Atomic Theory (1808) proposed that matter is composed of extremely small, indivisible particles called atoms, and that atoms are neither created nor destroyed during chemical reactions, but merely rearranged. This idea formed the conceptual basis of the Law of Conservation of Mass.
- **Statement 2 is incorrect:** Rutherford's nuclear model (1911), though revolutionary in establishing the existence of a dense central nucleus, failed to explain both atomic stability and the line spectrum of hydrogen. Under classical electromagnetic theory, orbiting electrons should continuously lose

energy, spiral into the nucleus, and collapse the atom. The model, therefore, could not account for discrete hydrogen spectral lines.

- **Statement 3 is correct:** Bohr's atomic model (1913) addressed Rutherford's limitations by proposing that electrons move only in certain fixed energy orbits, called stationary states, without radiating energy. Energy is emitted or absorbed only when electrons transition between these discrete levels, thereby explaining the hydrogen line spectrum.

Q 47. Ans. B

Explanation:

- **Law of Conservation of Mass**: Antoine Lavoisier established through careful experimental measurement that matter is neither created nor destroyed in chemical reactions, the total mass of reactants equals the total mass of products. This foundational principle became the basis of modern quantitative chemistry.
- **Discovery of electron:** Through his cathode ray tube experiments (1897), Thomson demonstrated the existence of negatively charged particles, later named electrons, with a mass far smaller than that of any known atom, establishing that atoms were not indivisible and possessed internal structure. He subsequently proposed the plum pudding model of the atom.
- **Discovery of neutron:** Chadwick demonstrated in 1932 the existence of electrically neutral particles within the atomic nucleus, which he called neutrons. This discovery explained the discrepancy between atomic mass and atomic number and completed the emerging modern understanding of atomic structure.
- **Discovery of proton:** While Rutherford is best known for the nuclear model of the atom (1911) derived from the gold foil experiment, he also identified the proton in 1919 as a positively charged nuclear particle through his transmutation experiments involving nitrogen bombardment with alpha particles.

Q 48. Ans. B

Explanation:

- **Statement 1 is correct:** Dalton's atomic theory postulated that all matter is composed of extremely small, indivisible particles called atoms, a foundational proposition that established the particulate nature of matter as the basis of chemical explanation.
- **Statement 2 is incorrect:** Dalton's theory explicitly stated that atoms of the same element are identical in mass and properties. The existence of isotopes, atoms of the same element with differing masses, was established much later through the work of Frederick Soddy and J.J. Thomson in the early 20th century, and directly contradicted Dalton's original postulate. Under Dalton's theory, variation in atomic mass within the same element was not recognised.
- **Statement 3 is correct:** A central postulate of Dalton's theory was that chemical reactions involve only the rearrangement, combination, or separation of atoms; atoms themselves are neither created nor destroyed in any chemical process. This directly supported the Law of Conservation of Mass.
- **Statement 4 is correct:** The Law of Definite Proportions, formulated by Joseph Proust (1799), states that a given chemical compound always contains the same elements united in a fixed proportion by mass. Dalton's atomic theory provided the theoretical explanation for this law, since atoms of given elements possess fixed masses, compounds formed from them must therefore exhibit constant mass ratios.

Q 49. Ans. A

Explanation:

- **Assertion (A) is true:** Neutrons have a mass approximately equal to that of a proton (~1.675 × 10^{2} kg, or approximately 1 atomic mass unit), making them significant contributors to the atomic mass of an element. The sum of protons and neutrons in the nucleus gives the mass number. Since neutrons carry no electrical charge, adding or removing neutrons from a nucleus changes the atomic mass but leaves the net charge of the atom entirely unaffected -- this is precisely why isotopes of the same element have identical chemical properties despite differing atomic masses.
- **Reason (R) is true:** Neutrons are electrically neutral particles, carrying zero charge and are located within the nucleus alongside protons. This was established by James Chadwick in 1932.
- **R correctly explains A:** The direct and complete reason why neutrons contribute to atomic mass without affecting net charge is precisely their nature as electrically neutral nuclear particles. Their location in the nucleus means they add to nuclear mass, and their electrical neutrality means they add nothing to the charge.

Q 50. Ans. B

Explanation:

- **Statement 1 is correct:** The atomic number is defined as the number of protons in the nucleus. With 19 protons, the atomic number is 19, corresponding to the element Potassium (K).
- **Statement 2 is correct:** The mass number is the sum of protons and neutrons in the nucleus. With 19 protons and 20 neutrons, the mass number is 19 + 20 = 39. This corresponds to Potassium-39 (^{39}K), the most abundant naturally occurring isotope of potassium.
- **Statement 3 is incorrect:** A positively charged ion (cation) is formed when an atom loses electrons, resulting in fewer electrons than protons. In this case, the atom has 19 protons and 19 electrons, equal numbers, meaning the positive nuclear charge is exactly balanced by the negative electronic charge, giving a net charge of zero. This is therefore a neutral atom, not a positively charged ion. A potassium cation (K^+) would have 19 protons but only 18 electrons.

Q 51. Ans. C

Explanation:

- **Statement 1 is correct:** The atomic number is defined solely by the number of protons in the nucleus. Losing an electron affects only the electron count, converting the atom into a positively charged cation, but leaves the number of protons, and therefore the atomic number and elemental identity, unchanged.
- **Statement 2 is incorrect:** The mass number is defined as the sum of protons and neutrons in the nucleus. Electrons have negligible mass (approximately 1/1836th of a proton's mass) and are not included in the mass number. Gaining an electron changes the ionic charge of the atom but does not alter the mass number. The atom becomes a negatively charged anion while retaining the same mass number.
- **Statement 3 is correct:** Since mass number = protons + neutrons, it follows that:
  - ✓ Number of neutrons = Mass number – Atomic number
  - ✓ This is the standard relation used to determine neutron count. For example, Potassium-39 has a mass number of 39 and an atomic number of 19, giving 39 – 19 = 20 neutrons.

Q 52. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Isotopes are atoms of the same element, having the same atomic number (same number of protons), but different mass numbers due to differing numbers of neutrons. For example, Carbon-12 (^{12}C) and Carbon-14 (^{14}C) both have an atomic number of 6 but different mass numbers. Isotopes generally exhibit identical chemical properties but different physical properties.
- **Pair 2 is correctly matched:** Isobars are atoms of different elements having the same mass number but different atomic numbers. For example, Calcium-40 (^{40}Ca, Z = 20) and Argon-40 (^{40}Ar, Z = 18) possess the same mass number but differ in proton count. Consequently, their chemical properties are different.
- **Pair 3 is correctly matched:** Isotones are atoms of different elements having the same number of neutrons but different atomic numbers and mass numbers. For example, Carbon-14 (6 protons, 8 neutrons) and Nitrogen-15 (7 protons, 8 neutrons) are isotones because both contain 8 neutrons.

Q 53. Ans. B

Explanation:

- **Statement 1 is correct:** Transuranic elements are, by definition, elements with atomic numbers greater than 92 (Uranium), the heaviest naturally occurring element in significant abundance. All elements beyond uranium in the periodic table are classified as transuranic, beginning with Neptunium (Z = 93).
- **Statement 2 is incorrect:** Transuranic elements do not occur naturally in significant abundance on Earth because their half-lives are generally far shorter than the age of the Earth, meaning any primordial quantities would have decayed long ago. Although trace amounts of certain transuranic elements such as neptunium and plutonium may arise naturally through nuclear processes in uranium ores, these quantities are negligible.
- **Statement 3 is correct:** Neptunium (Z = 93) was among the first artificially synthesised transuranic elements, produced in 1940 by Edwin McMillan and Philip Abelson through neutron bombardment of uranium. Plutonium (Z = 94) was synthesised shortly thereafter by Glenn Seaborg and collaborators. These discoveries marked the beginning of transuranic chemistry.

Q 54. Ans. A

Explanation:

- **Statement 1 is correct:** Sodium (Na) has an atomic number of 11 with electronic configuration 2, 8, 1, placing one electron in its outermost shell. This single valence electron is readily lost to form Na^+, giving sodium a valency of 1.
- **Statement 2 is correct:** Noble gases (Group 18) generally possess a stable octet, that is, 8 electrons in the outermost shell, which explains their chemical inertness. Helium is the exception, possessing a stable duplet (2 electrons) because the first shell is completely filled with two electrons.
- **Statement 3 is incorrect:** Valency is not always equal to the number of valence electrons. For elements with 1 to 4 valence electrons, valency generally equals the number of valence electrons. For elements with 5 to 7 valence electrons, valency is usually calculated as 8 minus the number of valence electrons, since gaining electrons is energetically more favourable than losing many electrons. For example, Nitrogen has 5 valence electrons but valency 3, while Oxygen has 6 valence electrons but valency 2.

Q 55. Ans. C

Explanation:

- **Statement 1 is correct:** Ionic bond formation involves the complete transfer of one or more electrons from an electropositive atom (typically a metal) to an electronegative atom (typically a non-metal), resulting in the formation of oppositely charged ions held together by electrostatic attraction. A classic example is NaCl, where sodium transfers one electron to chlorine, forming Na^+ and Cl^- ions.
- **Statement 2 is correct:** A covalent bond is formed by the mutual sharing of electron pairs between atoms, allowing them to attain stable electronic configurations. Examples include H_2, O_2, and CH_4.
- **Statement 3 is correct:** In a coordinate bond (dative bond), both electrons of the shared pair are contributed by one atom (the donor), while the other atom (the acceptor) provides an empty orbital. Examples include NH_4^+ (ammonium ion) and H_3O^+ (hydronium ion).
- **Statement 4 is incorrect:** Ionic compounds generally possess high melting and boiling points, not low ones, because strong electrostatic forces between oppositely charged ions in the crystal lattice require substantial energy to overcome. For example, NaCl melts at 801°C. Low melting points are more characteristic of many molecular covalent compounds.

Q 56. Ans. A

Explanation:

- Assertion (A) is true because an ammeter is used to measure the current flowing through a circuit, and therefore, the entire circuit current must pass through it. Hence, it is connected in series.
- Reason(R) is also true, as an ammeter is specially designed to have very low resistance. Because of this negligible resistance, the ammeter does not appreciably oppose or reduce the current flowing in the circuit. Thus, the current measured by the ammeter remains nearly equal to the actual circuit current.

Q 57. Ans. A

Explanation:

- Given:
  - ✓ Resistance of electric lamp, $R_1 = 20\ \Omega$
  - ✓ Resistance of conductor, $R_2 = 4\ \Omega$
  - ✓ Battery voltage, $V = 6$ V
- Since the lamp and conductor are connected in series, their resistances add up:
  - ✓ $R = R_1 + R_2$
  - ✓ $R = 20\ \Omega + 4\ \Omega$
  - ✓ $R = 24\ \Omega$
- Using Ohm's Law:
  - ✓ $I = V / R$
  - ✓ $I = 6\text{ V} / 24\ \Omega$
  - ✓ $I = 0.25$ A
  - ✓ Therefore, the current flowing through the circuit is **0.25 A.**
- Now, the potential difference across the electric lamp is:
  - ✓ $V_1 = I \times R_1$
  - ✓ $V_1 = 0.25\text{ A} \times 20\ \Omega$
  - ✓ $V_1 = 5$ V
- Thus, the potential difference across the electric lamp is 5 V.
- Hence the answer is **0.25 A, 5 V.**
- In a series circuit, the same current flows through all components, while the battery voltage is divided among them in proportion to their resistances. Since the lamp has a higher resistance (20 Ω) than the conductor (4 Ω), it receives a larger share of the total voltage (5 V out of 6 V).

Q 58. Ans. C

Explanation:

- **Statement 1 is correct:** Electric charges do not flow in a conductor by themselves. They move only when there is a difference in electric potential (potential difference) between two points of the conductor. This potential difference is generally produced by an electric cell or battery.
- **Statement 2 is correct:** The SI unit of electric potential difference is volt (V). One volt is defined as the potential difference between two points when 1 joule of work is done to move 1 coulomb of charge.

Q 59. Ans. A

Explanation:

- **Assertion (A) is true:** An electromagnet loses its magnetism as soon as the electric current flowing through it is switched off. This is because the magnetic field of an electromagnet is produced entirely by the flow of electric current through the coil wound around the core. When the current is interrupted, the magnetic field collapses immediately and the core, being made of soft iron, does not retain any significant residual magnetism. This property of losing magnetism upon the cessation of current is what distinguishes an electromagnet from a permanent magnet.
- **Reason (R) is true:** Electromagnets are indeed generally constructed by winding insulated copper wire around a soft iron core. Copper wire is used because copper is an excellent conductor of electricity and offers low resistance to the flow of current. The wire is insulated to prevent short circuits between adjacent turns of the coil. Soft iron is used as the core material because it is a highly permeable magnetic material that acquires magnetism quickly and strongly when placed within a magnetic field created by current carrying coils.
- **However, Reason (R) is not the correct explanation of Assertion (A):** The fact that an electromagnet loses its magnetism when current is switched off is explained by the temporary nature of the magnetism induced in the soft iron core by the electric current, and not merely by the construction materials used. The correct explanation lies in the property of soft iron, which has low retentivity, meaning it does not retain magnetism after the magnetising force is removed. The use of copper wire and soft iron core describes the construction of an electromagnet but does not explain why it loses its magnetism when the current is switched off. The

Q 60. Ans. B

Explanation:

- **Option A is incorrect:** Nichrome does not have low resistivity. It is specifically its **high** resistivity that makes it suitable for heating devices. Copper, which has low resistivity, is used for wiring, not heating.
- **Option B is correct:** Nichrome is an alloy of nickel and chromium. It has high electrical resistivity, which means it produces a large amount of heat when current passes through it. Additionally, it does not oxidise even at very high temperatures, making it ideal and durable for use in heating elements such as electric irons, heaters, toasters and immersion rods.
- **Option C is incorrect:** Nichrome does not show a significant decrease in resistance with temperature. This behaviour is characteristic of semiconductors, not of nichrome or metallic alloys.
- **Option D is incorrect:** Nichrome has a **high** melting point (approximately 1400°C), not a low one. This high melting point is precisely what makes it safe and reliable in high temperature heating applications. It is also not particularly malleable compared to pure metals.

Q 61. Ans. A

Explanation:

- **Statement 1 is correct:** A double bond consists of one σ bond and one π bond. The σ bond is formed by head-on overlap of orbitals, while the π bond is formed by lateral overlap of unhybridised p-orbitals. Therefore, a hydrocarbon containing one double bond, such as ethene (C_2H_4), contains exactly one π bond.
- **Statement 2 is correct:** A six-membered ring with alternating double bonds, benzene (C_6H_6), contains three π bonds in the conventional Kekulé representation, which is greater than a hydrocarbon containing a single double bond. These π electrons are delocalised over the ring, giving benzene its characteristic aromatic stability.
- **Statement 3 is incorrect:** A greater number of π bonds does not necessarily imply fewer σ bonds. σ bonds form the structural framework of molecules, since every single bond is a σ bond, and every double or triple bond contains one σ bond. A larger molecule may possess both more π bonds and more σ bonds.

For example, benzene has 3 π bonds and 12 σ bonds, whereas ethene has 1 π bond and 5 σ bonds.

Q 62. Ans. A

Explanation:

- Bond order is calculated using the formula:
  - ✓ Bond Order = (Number of bonding electrons – Number of antibonding electrons)/2
- For the bond order to equal zero, the numerator must be zero, meaning the number of bonding electrons must exactly equal the number of antibonding electrons. The stabilising effect of bonding electrons is completely cancelled by the destabilising effect of antibonding electrons, meaning no net bond exists and the diatomic species is unstable or non-existent. The classic example is He_2, which would have 2 bonding electrons (σ1s) and 2 antibonding electrons (σ^*1s), giving bond order = (2−2)/2 = 0, explaining why helium exists as a monoatomic gas rather than forming diatomic molecules.

Q 63. Ans. C

Explanation:

- Burning of LPG is a chemical change. Combustion of LPG (mainly propane and butane) produces carbon dioxide and water through oxidation, resulting in the formation of entirely new substances.
- Curdling of milk is a chemical change. Bacterial conversion of lactose into lactic acid alters the chemical composition of milk and causes protein coagulation, producing curd.
- Crushing of sugar crystals is a physical change. Only the size and shape of the sugar particles change; the substance remains sucrose ($C_{12}H_{22}O_{11}$) with no change in chemical composition.
- The setting of cement is a chemical change. Addition of water initiates hydration reactions that form new compounds, such as calcium silicate hydrate, producing a hardened structure with properties different from the original cement.

Q 64. Ans. A

Explanation:

- **Assertion (A) is true:** A catalyst increases the rate of both forward and backward reactions by providing an alternative pathway with lower activation energy. As a result, equilibrium is reached more quickly, but the equilibrium position and equilibrium constant remain unchanged.
- **Reason (R) is true:** A catalyst lowers the activation energy (E_a) required for the reaction to proceed, thereby increasing the reaction rate. It does not alter thermodynamic quantities such **as ΔG, ΔH, or the equilibrium constant (K).**
- **R correctly explains A:** Since the catalyst lowers the activation energy for both forward and reverse reactions without affecting the thermodynamic driving force, both reaction rates increase while the equilibrium position remains unchanged.

Q 65. Ans. C

Explanation:

- Photosynthesis is correctly matched as endothermic. Plants absorb solar energy to convert carbon dioxide and water into glucose and oxygen, storing absorbed energy in the chemical bonds of glucose.
- Melting of ice is endothermic, not exothermic. Heat must be absorbed from the surroundings to overcome intermolecular forces and convert solid ice into liquid water.
- Respiration is exothermic, not endothermic. The oxidation of glucose releases energy, which is utilised in ATP synthesis and metabolic activities.
- Neutralisation is exothermic. The reaction between an acid and a base releases energy, as $H^+ + OH^- \rightarrow H_2O$ is an energy-releasing process.

Q 66. Ans. B

Explanation:

- pH is defined as:
  - ✓ $pH = -\log_{10}[H^+]$
- Since pH is a logarithmic scale, a change of 1 pH unit corresponds to a 10-times change in hydrogen ion concentration.
- When pH increases from 3 to 5, the change is 2 units.
  - ✓ At pH 3: $[H^+] = 10^{-3}$ mol/L
  - ✓ At pH 5: $[H^+] = 10^{-5}$ mol/L
- Therefore:
  - ✓ $10^{-3} / 10^{-5} = 10^2 = 100$
- Thus, the hydrogen ion concentration decreases by 100 times.
- A higher pH indicates lower acidity and, therefore, lower hydrogen ion concentration.
- Correct answer: Decreases by 100 times

Q 67. Ans. A

Explanation:

- **Statement 1 is correct:** Phenolphthalein remains colourless in acidic and neutral media and turns pink in alkaline medium, making it an indicator for detecting alkalinity.

- **Statement 2 is correct:** Methyl orange appears red in acidic medium and yellow in neutral or alkaline medium, owing to its transition range in the acidic pH region.
- **Statement 3 is incorrect:** Dry blue litmus paper does not turn red in dry hydrogen chloride gas because HCl gas does not ionise in the absence of water. Litmus responds to hydrogen ions in an aqueous medium, so only moist blue litmus paper would turn red in the presence of HCl gas.

Q 68. Ans. B

Explanation:

- Distilled water exhibits extremely weak but measurable electrical conductivity due to the self-ionisation (autoionisation) of water:
  - ✓ $2H_2O \rightleftharpoons H_3O^+ + OH^-$
- At 25°C, the ionic product of water is:
  - ✓ $Kw = 1 \times 10^{-14}$
- Thus:
  - ✓ $[H^+] = [OH^-] = 10^{-7}$ mol/L
- Although this ion concentration is extremely small, the presence of these mobile ions allows distilled water to conduct electricity very weakly. Therefore, distilled water is a very poor conductor, though not a perfect insulator.

Q 69. Ans. A

Explanation:

- **Statement 1 is correct:** Molarity (M) is defined as the number of moles of solute per litre of solution:
  - ✓ M = moles of solute/volume of solution in litres
  - ✓ Since volume changes with temperature, molarity is temperature-dependent.
- **Statement 2 is incorrect:** The given definition describes normality, not molality. Molality (m) is defined as the number of moles of solute per kilogram of solvent:
  - ✓ m = moles of solute/mass of solvent in kg
  - ✓ Because it depends on mass rather than volume, molality is temperature-independent.
- **Statement 3 is incorrect:** The given definition corresponds to molarity, not normality. Normality (N) is defined as the number of gram equivalents of solute per litre of solution:
  - ✓ N = gram equivalents of solute/volume of solution in litres
  - ✓ Normality depends on the nature of the reaction involved.

Q 70. Ans. B

Explanation:

- **Option A is incorrect:** Poor electrical conductivity and brittleness alone describe non-metals rather than metalloids. Metalloids are specifically distinguished by their intermediate and conditional conductivity; silicon, for instance, is a semiconductor rather than an outright non-conductor.
- **Option B is correct:** Metalloids, also called semimetals, are elements that display a blend of both metallic and non-metallic properties, placing them along the "staircase" boundary on the periodic table. This intermediate character is their defining classification criterion. Examples include Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Boron (B), and Tellurium (Te). Their most technologically significant property is semiconductivity; they conduct electricity under certain conditions (elevated temperature or when doped) but not others, making them the foundation of modern electronics and solar cells.
- **Option C is incorrect:** Readily losing electrons to form only cations is characteristic of metals, particularly highly electropositive alkali and alkaline earth metals. Metalloids have intermediate electronegativity and can exhibit varied chemical behaviour depending on reaction conditions.
- **Option D is incorrect:** Existing exclusively in a gaseous state at room temperature is characteristic of noble gases and some non-metals like fluorine, chlorine, and nitrogen. All metalloids are solid at room temperature.

Q 71. Ans. C

Explanation:

- **Assertion (A) is true:** In thermodynamics, energy can be transferred to or from a system in two forms, heat and work. Heat transfer occurs due to a temperature difference, whereas work transfer occurs through mechanical means such as compression or expansion.
- **Reason (R) is false:** Internal energy includes the kinetic and potential energies of the molecules constituting the system, but it does not include the overall kinetic energy of the system as a whole.
- Hence, the assertion is true, but Reason is false. Therefore, the correct answer is Option C.

Q 72. Ans. A

Explanation:

- **Statement 1 is correct:** A radiation pyrometer is used to measure very high temperatures of distant bodies such as the Sun and stars without touching them. It estimates temperature through emitted radiation.

- **Statement 2 is correct:** A thermocouple works on the principle of the **Seebeck effect**, according to which an e.m.f. is generated when two dissimilar metals are joined, and their junctions are maintained at different temperatures.
- **Statement 3 is incorrect:** Thermo-electric e.m.f. is produced when **two dissimilar metals**, not identical metals, are joined together and subjected to a temperature difference.

Q 73. Ans. C

Explanation:

- **Pair 1 is correctly matched:** An open system can exchange both mass and energy with the surroundings. A water heater is a common example of an open system.
- **Pair 2 is correctly matched:** A closed system can exchange energy but not mass with the surroundings. Gas enclosed in a cylinder fitted with a piston is a standard example.
- **Pair 3 is correctly matched:** An isolated system exchanges neither mass nor energy with the surroundings. A filled thermos flask is considered an example.

Q 74. Ans. D

Explanation:

- The triple point of a substance refers to the unique condition of temperature and pressure at which all three physical states, solid, liquid and gas coexist simultaneously in equilibrium.
- On increasing pressure, the melting point of a solid decreases and the boiling point of the liquid increases. It is possible that by adjusting temperature and pressure, we can obtain all three states of matter to co-exist simultaneously. These temperature and pressure values signify the triple point.
- In the case of water, ice (solid), liquid water, and water vapour (gas) exist together at the triple point.

Q 75. Ans. C

Explanation:

- **Statement 1 is correct:** Conduction is the transfer of heat from a hotter region to a colder region without actual movement of matter. It is most effective in solids because particles are closely packed.
- **Statement 2 is correct:** Convection is the mode of heat transfer in liquids and gases through the actual movement of particles. Hotter, lighter fluid rises while colder, denser fluid sinks, forming convection currents.
- **Statement 3 is incorrect:** Radiation does not require any material medium. Heat from hot bodies can travel through a vacuum in the form of electromagnetic waves.
- **Statement 4 is correct:** Heat from the Sun reaches the Earth through radiation because the space between the Sun and Earth is largely a vacuum where conduction and convection cannot occur.

Q 76. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Zeroth Law establishes the concept of thermal equilibrium and temperature measurement.
- **Pair 2 is correctly matched:** The First Law of Thermodynamics expresses conservation of energy.
- **Pair 3 is correctly matched:** The Second Law states that heat naturally flows from a hot body to a cold body.
- **Pair 4 is incorrectly matched:** The entropy of a perfect crystal becomes minimum (zero) at absolute zero temperature. Hence, only pairs 1, 2 and 3 are correctly matched

Q 77. Ans. C

Explanation:

- Rock salt is sodium chloride (NaCl), composed exclusively of sodium and chlorine atoms, containing no oxygen in its chemical composition. It is an ionic compound formed by electrostatic attraction between Na^+ and Cl^- ions, with no oxygen-containing group present.
- Quartz is silicon dioxide (SiO_2), a three-dimensional covalent network solid in which each silicon atom is bonded to oxygen atoms. Limestone is calcium carbonate ($CaCO_3$), containing the carbonate ion (CO_3^{2-}) with three oxygen atoms per formula unit. Gypsum is calcium sulphate dihydrate ($CaSO_4 \cdot 2H_2O$), containing oxygen both in the sulphate group and in the water of crystallisation.
- A useful pattern is that compounds containing carbonate, sulphate, silicate, nitrate, phosphate, or water of crystallisation necessarily contain oxygen, whereas many halide salts such as chlorides, bromides, iodides, and fluorides do not.

Q 78. Ans. B

Explanation:

- **Statement 1 is incorrect:** Noble gases (Group 18) are chemically inert, not reactive. Their outermost shells are completely filled, helium has a stable duplet, while the others possess a complete octet, making them highly stable and minimally reactive.
- **Statement 2 is correct:** Halogens (Group 17, fluorine, chlorine, bromine, iodine, and astatine) have seven valence electrons and require only one additional electron to complete their outermost shell and achieve a stable octet configuration. This makes them highly electronegative and gives them a strong tendency to gain electrons, making them among the most reactive non-metals. Fluorine is the most reactive halogen and the most electronegative element.
- **Statement 3 is incorrect:** Bleaching powder is produced by passing chlorine gas over dry slaked lime [$Ca(OH)_2$]. Helium has no role in this process. Its applications lie in cryogenics, MRI systems, balloons, and shielding atmospheres.
- **Statement 4 is incorrect:** Chlorine belongs to Group 17 (halogens), not the noble gas group. It is highly reactive and readily forms compounds.

Q 79. Ans. D

Explanation:

- Glucose ($C_6H_{12}O_6$) and fructose ($C_6H_{12}O_6$) share the same molecular formula but differ in their structural arrangements, glucose is an aldohexose (containing an aldehyde group, –CHO) while fructose is a ketohexose (containing a ketone group, C=O). This makes them structural isomers, specifically functional group isomers. Despite having identical molecular formulae, they exhibit different chemical properties, glucose reacts directly with Fehling's solution, while fructose does so through tautomerisation.
- Isotopes are atoms of the same element with the same atomic number but different mass numbers due to differing neutron counts, a concept from atomic physics unrelated to molecular structure. Example: Carbon-12 and Carbon-14.
- Homologues are compounds belonging to the same homologous series, differing by a repeating $–CH_2–$ unit while sharing the same general formula and functional group. Example: methane (CH_4), ethane (C_2H_6), and propane (C_3H_8).
- Polymers are large macromolecules formed by the repetitive joining of smaller monomer units through covalent bonds. Example: starch and cellulose are polymers of glucose.

Q 80. Ans. A

Explanation:

- Urea (NH_2CONH_2), also written as $CO(NH_2)_2$, contains carbon, hydrogen, nitrogen, and oxygen, with the molecular formula CH_4N_2O. Its structure consists of a central carbonyl group (C=O) flanked by two amino groups ($–NH_2$), making it the diamide of carbonic acid.
- Urea occupies a landmark position in the history of chemistry. It was first synthesised by Friedrich Wöhler in 1828 by heating ammonium cyanate (NH_4OCN), an inorganic compound. This was the first laboratory synthesis of an organic compound from an inorganic precursor, decisively challenging the vital force theory, which had claimed that organic compounds could only be produced by living organisms. Wöhler's experiment is regarded as a foundational moment in modern organic chemistry.
- Biologically, urea is the principal nitrogenous waste product of protein metabolism in mammals, produced in the urea cycle in the liver and excreted through urine. Industrially, it is manufactured by reacting ammonia with carbon dioxide under high temperature and pressure, and is the world's most widely used nitrogen fertiliser, containing approximately 46% nitrogen by mass.

Q 81. Ans. A

Explanation:

- **Statement A is correct:** The extraordinary diversity of organic compounds, currently numbering over 10 million known compounds, stems directly from carbon's unique combination of tetravalency (forming four covalent bonds) and catenation (forming stable bonds with other carbon atoms). Carbon can bond with hydrogen, oxygen, nitrogen, sulphur, halogens, and phosphorus while simultaneously forming chains, branches, and rings of immense structural complexity. This versatility underlies the entire field of organic chemistry.
- **Statement B is incorrect:** Organic compounds are characterised by remarkable structural complexity and diversity, not simplicity. A single molecular formula may correspond to multiple structural isomers. For example, C_5H_{12} has three structural isomers, while larger molecules exhibit even greater isomeric diversity.

- **Statement C is incorrect:** Not all carbon-containing compounds are organic. Carbon dioxide (CO_2), carbon monoxide (CO), carbonates, bicarbonates, and cyanides are conventionally classified as inorganic compounds despite containing carbon.
- **Statement D is incorrect:** Organic compounds are predominantly characterised by covalent bonding, not ionic bonding. This accounts for many of their typical properties, including comparatively lower melting and boiling points and distinctive solubility behaviour.

Q 82. Ans. B

Explanation:

- **Zymase** is the enzyme complex responsible for converting glucose directly into ethyl alcohol (ethanol) and carbon dioxide through the process of anaerobic fermentation:
  - ✓ $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- **Zymase** is present in yeast (Saccharomyces cerevisiae) and functions optimally at around 30--35°C. It is actually a complex of multiple enzymes working sequentially through the glycolytic pathway, collectively catalysing the breakdown of glucose to pyruvate and subsequently to ethanol. This process forms the biochemical basis of the production of beer, wine, whisky, and industrial ethanol. In a broader fermentation sequence, another enzyme invertase (also present in yeast) first converts sucrose into glucose and fructose, which zymase then ferments.
- **Pepsin** is a proteolytic enzyme secreted by the gastric glands of the stomach in its inactive form pepsinogen, which is activated by hydrochloric acid. It catalyses the hydrolysis of proteins into smaller peptides (proteolysis) and functions optimally in the strongly acidic environment of the stomach (pH ~2). It is entirely unrelated to carbohydrate fermentation.
- **Maltase** is a digestive enzyme found in the small intestine and in yeast that catalyses the hydrolysis of maltose (a disaccharide) into two molecules of glucose. It acts on the products of starch digestion by amylase, completing the breakdown of disaccharides into absorbable monosaccharides. It is a hydrolytic enzyme, not a fermentation enzyme.
- **Trypsin** is a serine protease produced by the pancreas as its inactive precursor trypsinogen, activated in the small intestine by the enzyme enterokinase. It hydrolyses peptide bonds in proteins and polypeptides, specifically cleaving on the carboxyl side of lysine and arginine residues. Like pepsin, it is a protein-digesting enzyme with no role in carbohydrate fermentation.

Q 83. Ans. B

Explanation:

- Incomplete combustion occurs when there is an insufficient supply of oxygen relative to the fuel being burned, a condition common in internal combustion engines, particularly during cold starts, idling, and acceleration. Under these conditions, carbon in the fuel is only partially oxidised, producing carbon monoxide (CO) instead of the fully oxidised product carbon dioxide (CO_2):
  - ✓ $2C + O_2 \rightarrow 2CO$
- CO is an extremely dangerous, colourless, and odourless gas. Its toxicity arises from its ability to bind to haemoglobin with approximately 250 times greater affinity than oxygen, forming carboxyhaemoglobin (COHb) and severely reducing the blood's oxygen-carrying capacity, causing symptoms ranging from headache and dizziness to death at high concentrations.
- Carbon dioxide (CO_2) is the product of complete combustion, formed when sufficient oxygen is available.
- Sulphur dioxide (SO_2) is produced when sulphur-containing fuels are burned and is associated with acid rain, not specifically incomplete combustion.
- Methane (CH_4) is a hydrocarbon fuel itself, not a combustion product of petrol or diesel.

Q 84. Ans. C

Explanation:

- Natural rubber is a naturally occurring polymer of isoprene (2-methyl-1,3-butadiene, C_5H_8), produced as a milky white latex by the rubber tree Hevea brasiliensis. The polymer chain consists of thousands of isoprene units linked together in a cis-1,4 configuration, giving natural rubber its characteristic elasticity. The systematic name of natural rubber is cis-polyisoprene. When the same isoprene units link in a trans-1,4 configuration, the result is gutta-percha, a naturally occurring hard, non-elastic material historically used in golf balls and dental fillings. Raw natural rubber is vulcanised by heating with sulphur to introduce cross-links between polymer chains, significantly improving its strength, elasticity, and durability for industrial applications.
- Ethylene ($CH_2{=}CH_2$) is the monomer of

polyethylene (polythene), one of the world's most widely produced synthetic plastics, used in packaging, bottles, and pipes. It has no relation to natural rubber.

- Styrene ($C_6H_5CH=CH_2$) is the monomer of polystyrene, a rigid synthetic polymer used in disposable cups, insulation, and packaging foam. When copolymerised with butadiene, it forms styrene-butadiene rubber (SBR), a widely used synthetic rubber substitute.
- Vinyl chloride ($CH_2=CHCl$) is the monomer of polyvinyl chloride (PVC), one of the most produced synthetic plastics globally, used in pipes, cables, flooring, and medical devices. It contains chlorine, entirely unlike the hydrocarbon structure of natural rubber.

Q 85. Ans. B

Explanation:

- Bakelite (polyoxybenzylmethylenglycolanhydride) is an example of a thermosetting polymer, synthesised by the condensation polymerisation of phenol and formaldehyde, first developed by Leo Baekeland in 1907, making it the world's first fully synthetic plastic. During its initial moulding under heat and pressure, Bakelite undergoes irreversible cross-linking between polymer chains, forming a rigid, three-dimensional network structure. Once set, it cannot be remelted, resoftened, or reshaped by subsequent heating, any attempt to reheat it results in charring or decomposition rather than softening. This thermal permanence makes it ideal for electrical switches, plugs, sockets, cookware handles, and automobile parts where heat resistance is essential.
- Thermoplastic polymers soften and become mouldable upon heating and re-harden upon cooling, a reversible process that can be repeated multiple times. They lack the permanent cross-linked structure of thermosets. Examples include polyethylene, PVC, nylon, and polystyrene.
- Biodegradable polymers are designed to decompose through microbial action in the environment, helping address plastic waste concerns. Examples include PLA (polylactic acid), PHBV, and natural polymers such as starch and cellulose.
- Natural polymers occur in living organisms without synthetic processing. Examples include natural rubber (cis-polyisoprene), cellulose, starch, proteins (silk, wool), and nucleic acids (DNA, RNA).

Q 86. Ans. A

Explanation:

- PHBV stands for Poly-β-hydroxybutyrate-co-β-hydroxyvalerate, a naturally occurring biodegradable copolymer belonging to the polyhydroxyalkanoate (PHA) family. It is produced by certain bacteria, notably Ralstonia eutropha, as an intracellular carbon and energy storage material under nutrient-limiting conditions.
- PHBV is formed by the copolymerisation of two monomeric units, β-hydroxybutyrate and β-hydroxyvalerate, and the relative proportion of these units determines the polymer's physical properties, especially flexibility, toughness, and melting point. A higher β-hydroxyvalerate content generally makes the polymer more flexible and less brittle.
- PHBV is classified as both a biopolymer (because it is produced by living organisms) and a biodegradable polymer. It decomposes through microbial action in soil and aquatic environments into carbon dioxide and water, without leaving toxic residues, making it an environmentally important alternative to conventional petroleum-based plastics.
- Its applications include biodegradable packaging, controlled drug delivery systems, surgical sutures, and tissue engineering scaffolds.

Q 87. Ans. B

Explanation:

- Nylon is a fully synthetic polyamide fibre, first developed by Wallace Carothers at DuPont in 1935, and is renowned for its exceptionally high tensile strength, elasticity, abrasion resistance, and durability. The most common variety, Nylon-6,6, is produced by the condensation polymerisation of hexamethylenediamine and adipic acid. It is stronger than steel wire of the same diameter on a weight-for-weight basis, making it suitable for parachutes, ropes, toothbrush bristles, stockings, fishing nets, and industrial gears. Its high strength-to-weight ratio and resistance to heat and chemicals make it one of the most versatile synthetic fibres ever produced.
- Rayon is a semi-synthetic fibre, produced from regenerated cellulose derived from wood pulp, but requiring extensive chemical processing involving dissolution in sodium hydroxide and carbon disulphide to form viscose solution before being reconstituted into fibres. It is neither fully natural nor fully synthetic.

- Acrylic is a fully synthetic fibre produced by the polymerisation of acrylonitrile (CH_2=CHCN), a petroleum-derived monomer. It resembles wool in texture and is widely used as a wool substitute in sweaters, blankets, carpets, and upholstery.
- Polyester, most commonly PET (polyethylene terephthalate), is a fully synthetic fibre produced by the condensation polymerisation of ethylene glycol and terephthalic acid, both petroleum derivatives. Animal-derived fibres include silk and wool, not polyester.

Q 88. Ans. A

Explanation:

- The assertion is correct because sound waves in gases travel through compressions and rarefactions, which are characteristic of longitudinal waves.
- The reason is also correct because the gas particles move back and forth parallel to the direction in which the wave travels. This directly explains why sound waves in gases are longitudinal.

Q 89. Ans. C

Explanation:

- Light waves and radio waves are electromagnetic waves and do not require a material medium for propagation. Therefore, they can travel through a vacuum.
- Sound waves are mechanical waves and require a material medium such as air, liquid, or solid for propagation. Hence, they cannot travel in a vacuum.

Q 90. Ans. B

Explanation:

- **Statement 1 is correct:** The Doppler Effect is the apparent change in frequency or pitch due to relative motion between the source and observer.
- **Statement 2 is incorrect:** When the source moves towards the observer, the observed frequency increases because the wavefronts get compressed.
- **Statement 3 is correct:** Sound waves commonly exhibit the Doppler Effect, such as the changing pitch of a train whistle or ambulance siren.
- **Statement 4 is incorrect:** Doppler Effect changes the observed frequency and wavelength, but the speed of sound in the medium remains unchanged.

Q 91. Ans. C

Explanation:

- The SI unit of frequency is Hertz (Hz).
- The SI unit of time period is second (s).
- The SI unit of amplitude is metre (m), not joule.
- The SI unit of wavelength is the metre (m).

Q 92. Ans. C

Explanation:

- **Statement 1 is correct:** In a harmonic progressive wave, all particles possess the same amplitude, but their phases differ from point to point at a given instant.
- **Statement 2 is also correct:** In a stationary wave, particles between two consecutive nodes vibrate in the same phase, but their amplitudes vary from zero at nodes to a maximum at antinodes.

Q 93. Ans. B

Explanation:

- **SONAR** is a technique that uses ultrasonic sound waves to detect, locate, and measure the distance of underwater objects.
- It is widely used in ships, submarines, oceanographic studies, and the fishing industry.
- **SONAR** works on the principle of reflection of sound waves (echoes).
  - ✓ A transmitter emits ultrasonic waves through water.
  - ✓ These waves strike an underwater object and get reflected.
  - ✓ The reflected waves (echoes) are received by a detector.
  - ✓ By measuring the time taken for the echo to return, the distance of the object is calculated.

Q 94. Ans. B

Explanation:

- The human ear retains the sensation of sound for about 0.1 second due to persistence of hearing. Therefore, a reflected sound is heard separately as an echo only if it reaches the ear after at least 0.1 seconds from the original sound.
- If the reflected sound arrives earlier, it merges with the original sound and cannot be distinguished separately.

Q 95. Ans. B

Explanation:

- **Statement 1 is correct:** The speed of longitudinal waves in a fluid depends upon two important properties of the medium, that is, the bulk **modulus (B), which** measures the elasticity or resistance to compression of the medium and the **density.**
- **Statement 2 is correct:** Bulk modulus is defined as

the ratio of the change in pressure to the fractional change in volume produced. It indicates how difficult it is to compress a substance. A large bulk modulus means the substance is less compressible and more rigid. Since sound waves propagate through successive compressions and rarefactions, a higher bulk modulus results in faster transmission of sound.

- **Statement 3 is incorrect:** Although solids possess much greater density than gases, the speed of sound in solids is still much higher. This is because the effect of elasticity is far more significant than density in determining wave speed.
  - ✓ Solids are highly rigid and much less compressible than gases. Their elastic modulus is extremely large, which greatly increases the speed of sound.
  - ✓ Typical speeds of sound:
    - **Air:** nearly 340 m/s
    - **Water:** nearly 1500 m/s
    - **Steel:** nearly 5000 m/s
- **Statement 4 is correct:** In solids such as rods or bars, longitudinal waves involve stretching and compression along the length of the solid. Therefore, the relevant elastic property is **Young's modulus (Y)** rather than the bulk modulus.

Q 96. Ans. A

Explanation:

- René Descartes (1637) proposed the corpuscular model of light and Isaac Newton later expanded this model in his book Opticks (1704).
- Christiaan Huygens (1690) gave the wave theory of light.
- Thomas Young (1801) is associated with the famous interference experiment of 1801.
- James Clerk Maxwell (1865) proposed the electromagnetic theory of light.

Q 97. Ans. D

Explanation:

- **Statement 1 is correct:** A wavefront is defined as the locus of all points vibrating in the same phase at a given instant. Hence, it is called a surface of constant phase.
- **Statement 2 is correct:** Wave energy propagates in a direction normal (perpendicular) to the wavefront. This direction is represented by rays.
- **Statement 3 is correct:** When a point source emits waves uniformly in all directions, the wavefront formed is spherical in shape.
- **Statement 4 is correct:** According to Huygens' principle, every point on a wavefront behaves as a source of secondary wavelets spreading in all directions.

Q 98. Ans. A

Explanation:

- **Assertion (A) is true:** At sufficiently large distances from the source, the curvature of the spherical wavefront becomes very small.
- **Reason (R) is also true:** A tiny section of a very large sphere appears almost flat, just as a small part of Earth's surface appears plane. Therefore, the reason correctly explains the assertion.

Q 99. Ans. B

Explanation:

- **Statement 1 is correct:** Huygens' principle explains refraction based on secondary wavelets. According to the Principle, "each point of the wavefront is the source of a secondary disturbance and the wavelets emanating from these points spread out in all directions with the speed of the wave.
- **Statement 2 is correct:** This means when light enters an optically denser medium, its velocity decreases. Since the refracted ray bends towards the normal, the angle of refraction becomes smaller than the angle of incidence.
- **Statement 3 is incorrect:** The frequency of light does not change during refraction because frequency is determined by the source of light and must remain continuous across the boundary.
- **Statement 4 is correct:** When light enters a denser medium, its speed decreases. Since frequency remains unchanged, wavelength also decreases proportionally.

Q 100. Ans. B

Explanation:

- **Option A is incorrect:** Light actually speeds up when entering a rarer medium, not slows down. Total internal reflection is not related to energy conservation through reflection. It occurs purely due to the geometry of refraction at the boundary.
- **Option B is correct:** When light travels from a denser medium to a rarer medium, it bends away from the normal upon refraction. As the angle of incidence increases, the angle of refraction also increases. At a specific angle of incidence called the critical angle, the angle of refraction reaches exactly 90°, meaning the refracted ray grazes the boundary

surface. Beyond this critical angle, refraction is no longer possible and the light is completely reflected back into the denser medium. This phenomenon is total internal reflection.

- **Option C is incorrect:** This describes the opposite of what actually happens. When light enters a rarer medium, it bends away from the normal, not towards it. Bending towards the normal occurs when light enters a denser medium.
- **Option D is incorrect:** Frequency of light never changes at a boundary, regardless of the media involved. Frequency is determined solely by the source. It is the wavelength and speed that change during refraction, not frequency.

Q 101. Ans. C

Explanation:

- In the photon model, light consists of discrete packets of energy called photons. The intensity of light depends on how many photons strike a unit area in a given time interval.
- The greater the number of photons crossing a unit area per unit time, the greater is the intensity of light. The energy of an individual photon is given by:
  - ✓ **E – hv**
- **Thus, frequency** determines the energy of each photon, and the number of photons determines the intensity of light.

Q 102. Ans. C

Explanation:

- Linen is a natural textile fibre obtained from the stem of the flax plant (Linum usitatissimum), one of the oldest cultivated plants in human history. The fibre is extracted through retting (soaking stems in water to loosen fibres), scutching (beating), and hackling (combing) before being spun into thread. Linen is valued for its high tensile strength, breathability, moisture-wicking properties, and natural lustre, making it ideal for summer clothing, bed linen, and high-quality writing paper.
- Plant-based (cellulosic) fibres:
  - ✓ Cotton – seed hair of the cotton plant (Gossypium)
  - ✓ Linen – stem of the flax plant (Linum usitatissimum)
  - ✓ Jute – stem of Corchorus plants; used in sacking, ropes, and gunny bags
  - ✓ Hemp – stem of Cannabis sativa; used in ropes, canvas, and industrial textiles
  - ✓ Coir – husk of coconut (Cocos nucifera); used in doormats, ropes, and mattresses
  - ✓ Sisal – leaves of Agave sisalana; used in ropes and twines
  - ✓ Ramie – stem of Boehmeria nivea; among the strongest natural fibres
- Animal-based (protein) fibres:
  - ✓ Wool – fleece of sheep (Ovis aries); composed of keratin protein
  - ✓ Silk – cocoon of silkworm (Bombyx mori); produced through sericulture
  - ✓ Cashmere – undercoat of Cashmere goat (Capra hircus)
  - ✓ Angora – fur of Angora rabbit or Angora goat (mohair)
  - ✓ Alpaca – fleece of alpaca (Vicugna pacos)
  - ✓ Pashmina – fine undercoat of Changthangi goat from the Ladakh region
  - ✓ Duck feathers are used as insulating fill material in pillows and duvets but are not conventional spun textile fibres, belonging instead to the category of down insulation.

Q 103. Ans. B

Explanation:

- **Option 1 is incorrect:** PTFE is actually a very poor conductor of both heat and electricity, which is why it is also used as an electrical insulator in wires and cables. Heat conduction in cookware comes from the underlying metal base (aluminium or steel), not the Teflon coating.
- **Option 2 is correct:** Teflon, chemically known as polytetrafluoroethylene (PTFE), is a synthetic fluoropolymer produced by the addition polymerisation of tetrafluoroethylene ($CF_2{=}CF_2$). Its suitability as a non-stick cookware coating arises from a unique combination of properties rooted directly in its molecular structure.
  - ✓ The carbon-fluorine (C–F) bond is one of the strongest bonds in organic chemistry (bond energy approximately 485 kJ/mol), making PTFE extraordinarily resistant to chemical attack by acids, bases, solvents, and food substances at cooking temperatures. This exceptional chemical inertness ensures that food does not react with or adhere chemically to the surface.
  - ✓ Additionally, PTFE remains thermally stable up to approximately 260°C under normal cooking conditions, retaining its structural integrity and non-stick behaviour without decomposition. Its extremely low surface energy and coefficient of friction, among the lowest of any solid material,

physically prevent food molecules from adhering to the surface, which is the direct basis of its non-stick character.

- **Option 3 is incorrect:** PTFE is not biodegradable. The exceptional stability of the carbon-fluorine bond makes it highly resistant to microbial degradation, causing it to persist in the environment for very long periods.
- **Option 4 is incorrect:** PTFE does not simply soften and spread under ordinary cooking conditions. Its application as a cookware coating requires specialised industrial processing, including surface preparation and high-temperature sintering, not simple thermal melting during use.

Q 104. Ans. C

Explanation:

- Saponification is the chemical process by which fats or oils (triglycerides) are heated with a strong alkali, typically sodium hydroxide (NaOH) for hard soaps or potassium hydroxide (KOH) for soft or liquid soaps, to produce soap (fatty acid salts) and glycerol as a byproduct. The reaction can be represented as:
  - ✓ Fat/Oil (Triglyceride) + NaOH → Soap (Sodium salt of fatty acid) + Glycerol
- The soap molecule possesses a characteristic dual structure, a hydrophilic (water-attracting) ionic head ($-COO^-Na^+$) and a hydrophobic (water-repelling) long hydrocarbon tail, enabling it to emulsify grease and oil through micelle formation during cleaning.
- Vulcanisation is the process of heating natural rubber with sulphur to introduce cross-links between polymer chains, thereby improving strength, elasticity, and durability. It has no connection with fat chemistry.
- Polymerisation is the process of linking small monomer molecules into large polymer chains through covalent bonding. Examples include ethylene to polyethylene and styrene to polystyrene.
- Hydrogenation involves the addition of hydrogen across unsaturated carbon-carbon double bonds in the presence of catalysts such as nickel, palladium, or platinum. It is used to convert vegetable oils into vanaspati.

Q 105. Ans. B

Explanation:

- Nitrous oxide (N_2O), commonly called laughing gas, is a colourless, sweet-smelling gas that produces euphoria, mild hallucination, and analgesic effects when inhaled in controlled quantities, hence its historical use as an anaesthetic, particularly in dentistry and minor surgical procedures. It was first synthesised by Joseph Priestley in 1772, and its anaesthetic properties were demonstrated by Humphry Davy in 1800. Today, it is also used as a propellant in whipped cream dispensers and as an oxidiser in motorsports engines.
- Several gases and compounds have acquired common or colloquial names distinct from their systematic chemical names:
- Nitrogen-based compounds:
  - ✓ Laughing gas – Nitrous oxide (N_2O)
  - ✓ Nitrogen dioxide (NO_2) – toxic reddish-brown gas contributing to smog and acid rain
  - ✓ Dinitrogen pentoxide (N_2O_5) – anhydride of nitric acid, powerful oxidising agent
- Oxides and acids:
  - ✓ Dry ice – Solid carbon dioxide (CO_2)
  - ✓ Oil of vitriol – Concentrated sulphuric acid (H_2SO_4)
  - ✓ Muriatic acid – Hydrochloric acid (HCl)
  - ✓ Aqua regia – Mixture of concentrated HCl and HNO_3 (3:1); dissolves gold and platinum
  - ✓ Spirit of salt – Hydrochloric acid (HCl)
  - ✓ Oleum – $H_2S_2O_7$ (fuming sulphuric acid)
- Common alkalis and salts:
  - ✓ Caustic soda – Sodium hydroxide (NaOH)
  - ✓ Caustic potash – Potassium hydroxide (KOH)
  - ✓ Slaked lime – Calcium hydroxide ($Ca(OH)_2$)
  - ✓ Quicklime – Calcium oxide (CaO)
  - ✓ Washing soda – Sodium carbonate decahydrate ($Na_2CO_3 \cdot 10H_2O$)
  - ✓ Baking soda – Sodium bicarbonate ($NaHCO_3$)
  - ✓ Milk of magnesia – Magnesium hydroxide ($Mg(OH)_2$)
  - ✓ Blue vitriol – Copper sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$)
  - ✓ Green vitriol – Ferrous sulphate heptahydrate ($FeSO_4 \cdot 7H_2O$)
  - ✓ White vitriol – Zinc sulphate heptahydrate ($ZnSO_4 \cdot 7H_2O$)
  - ✓ Epsom salt – Magnesium sulphate heptahydrate ($MgSO_4 \cdot 7H_2O$)
  - ✓ Gypsum – Calcium sulphate dihydrate ($CaSO_4 \cdot 2H_2O$)
  - ✓ Plaster of Paris – Calcium sulphate hemihydrate ($CaSO_4 \cdot ½H_2O$)
- Organic compounds:
  - ✓ Wood alcohol / Wood spirit – Methanol (CH_3OH)

- ✓ Grain alcohol – Ethanol (C_2H_5OH)
- ✓ Acetone – Propanone (CH_3COCH_3)
- ✓ Marsh gas – Methane (CH_4)
- ✓ Black damp / Choke damp – Carbon dioxide (CO_2)
- ✓ Firedamp – Methane (CH_4) found in coal mines
- ✓ Chloroform – Trichloromethane ($CHCl_3$)
- ✓ Carbolic acid – Phenol (C_6H_5OH)
- ✓ Formic acid – Methanoic acid (HCOOH)
- ✓ Acetic acid – Ethanoic acid (CH_3COOH)

Q 106. Ans. B

Explanation:

- Potassium dichromate ($K_2Cr_2O_7$) is the chemical agent used in traditional breathalyzer tests to detect ethanol (alcohol) in exhaled breath. The test is based on a characteristic redox reaction in which ethanol is oxidised to acetic acid, while the dichromate ion ($Cr_2O_7^{2-}$) is simultaneously reduced:
- $3C_2H_5OH + K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow 3CH_3COOH + Cr_2(SO_4)_3 + K_2SO_4 + 7H_2O$
- This reaction produces a clear colour change from orange (dichromate ion) to green (Cr^{3+} ion), the intensity of which is proportional to the alcohol concentration in the breath, and therefore correlates with blood alcohol content.
- This colour-change principle formed the basis of traditional chemical breathalyzers, although many modern devices now use infrared spectroscopy or electrochemical fuel cell sensors for greater precision.
- Potassium permanganate ($KMnO_4$), though also a strong oxidising agent, is not the standard reagent used in traditional breathalyzer chemistry.
- Copper sulphate ($CuSO_4$) is used in tests such as Fehling's and Benedict's tests for reducing substances.
- Sodium bicarbonate ($NaHCO_3$) has no role in alcohol detection.

Q 107. Ans. D

Explanation:

- Biogas is produced by the anaerobic digestion of organic waste materials, including animal dung, agricultural residues, food waste, and sewage sludge, by methanogenic bacteria in the absence of oxygen. Its typical composition is 55–70% methane (CH_4), 30–40% carbon dioxide (CO_2), with trace amounts of hydrogen sulphide, nitrogen, and water vapour. Methane (CH_4) is the chief combustible constituent, burning with a clean blue flame and releasing substantial energy. India's National Biogas and Manure Management Programme (NBMMP) promotes biogas as a clean rural cooking fuel.
- Gaseous fuels:
  - ✓ Biogas – chief constituent methane (CH_4), 55–70%
  - ✓ Natural gas – chiefly methane (CH_4), up to 95%
  - ✓ LPG (Liquefied Petroleum Gas) – mainly propane (C_3H_8) and butane (C_4H_{10})
  - ✓ CNG (Compressed Natural Gas) – primarily methane (CH_4)
  - ✓ Coal gas (Town gas) – mixture of hydrogen, methane, and carbon monoxide
  - ✓ Water gas (Syngas) – mixture of carbon monoxide and hydrogen
  - ✓ Producer gas – mixture of carbon monoxide and nitrogen
  - ✓ Hydrogen gas – pure H_2
- Liquid fuels:
  - ✓ Petrol – mixture of C_5–C_{12} hydrocarbons
  - ✓ Diesel – mixture of C_{12}–C_{25} hydrocarbons
  - ✓ Kerosene – mixture of C_{10}–C_{16} hydrocarbons
  - ✓ Ethanol – C_2H_5OH
  - ✓ Methanol – CH_3OH
- Solid fuels:
  - ✓ Coal – chiefly carbon
  - ✓ Coke – nearly pure carbon
  - ✓ Wood/Biomass – cellulose and lignin
  - ✓ Charcoal – porous carbon
- Non-combustible components often present in fuels:
  - ✓ CO_2 – non-combustible diluent
  - ✓ H_2S – toxic impurity
  - ✓ N_2 – inert diluent

Q 108. Ans. C

Explanation:

- When carbon dioxide is passed through limewater, a dilute aqueous solution of calcium hydroxide $Ca(OH)_2$, the following reaction occurs:
  - ✓ $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3\downarrow + H_2O$
- The milky white precipitate formed is calcium carbonate ($CaCO_3$), which is insoluble in water and immediately appears as a white suspension, giving limewater its characteristic turbidity. This reaction is the standard laboratory test for carbon dioxide, a gas is confirmed to be CO_2 if it turns limewater milky.
- An important extension of this reaction occurs

when excess CO_2 is passed through the already milky limewater:

✓ $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

- The milkiness disappears as the insoluble calcium carbonate is converted into calcium bicarbonate $Ca(HCO_3)_2$, which is soluble in water, producing a clear solution again. If this clear solution is subsequently heated, the reaction reverses, calcium bicarbonate decomposes back to insoluble calcium carbonate, and the milkiness reappears. This reversible sequence is also the chemical basis of cave formation, slightly acidic rainwater containing dissolved CO_2 dissolves limestone ($CaCO_3$) to form soluble $Ca(HCO_3)_2$, which redeposits as $CaCO_3$ to form stalactites and stalagmites.
- Calcium oxide (CaO) is quicklime, produced by strongly heating limestone, not by the reaction of CO_2 with limewater. Gaseous calcium hydroxide is chemically meaningless under normal conditions, as $Ca(OH)_2$ is a solid ionic compound. Calcium bicarbonate is formed only upon passing excess CO_2 and is soluble, it causes the disappearance of milkiness rather than its appearance.

Q 109. Ans. B

Explanation:

- Calcium carbonate reacting with dilute hydrochloric acid produces carbon dioxide gas through the following reaction:

  ✓ $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2\uparrow$
- This is a acid-carbonate reaction, whenever any carbonate or bicarbonate reacts with an acid, CO_2 is invariably evolved. The reaction is readily identifiable by the effervescence produced and can be confirmed by passing the evolved gas through limewater, which turns milky. This reaction is the standard laboratory method for preparing CO_2 and also explains the fizzing observed when baking soda reacts with vinegar.
- Sodium chloride reacting with hydrochloric acid produces no gas, NaCl and HCl are both chloride compounds and their reaction yields no new products under normal conditions. This is essentially a non-reaction as no driving force exists.
- Copper reacting with dilute sulphuric acid produces no reaction under normal conditions. Copper lies below hydrogen in the reactivity series and cannot displace hydrogen from dilute acids. Only concentrated hot sulphuric acid reacts with copper, producing sulphur dioxide (SO_2), not CO_2.
- Zinc reacting with sodium hydroxide solution produces hydrogen gas (H_2), not CO_2. Zinc is an amphoteric metal that reacts with alkalis:

  ✓ $Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2\uparrow$
- This reaction is notable as most metals do not react with alkalis, zinc's amphoteric nature makes it an exception.

Q 110. Ans. A

Explanation:

- Tetraethyl lead (TEL, $Pb(C_2H_5)_4$) was historically added to petrol in small quantities as an anti-knock additive to improve its octane rating, a measure of a fuel's resistance to premature ignition (knocking) in internal combustion engines. Knocking occurs when the fuel-air mixture in a cylinder ignites before the spark plug fires, causing a characteristic metallic rattling sound and reducing engine efficiency and longevity. TEL suppresses this premature ignition by interrupting the free radical chain reactions responsible for autoignition, allowing the fuel to burn smoothly and controllably.
- TEL was first introduced commercially by General Motors in 1923, developed by Thomas Midgley Jr. Its use allowed engine compression ratios to be raised significantly, improving power output and fuel efficiency. However, the lead compounds (lead oxide and lead bromide) produced during combustion were released into the atmosphere as toxic exhaust emissions, causing severe and widespread lead poisoning, particularly affecting neurological development in children. Mounting scientific evidence linking leaded petrol to elevated blood lead levels in urban populations led to its global phaseout through the later decades of the 20th century. India banned leaded petrol in 2000, and the United Nations Environment Programme (UNEP) declared the world free of leaded petrol for road vehicles in 2021, when Algeria, the last country using it, completed its phaseout.

Q 111. Ans. D

Explanation:

- Portland cement is manufactured by heating a precisely controlled mixture of limestone ($CaCO_3$) and clay (containing silica, alumina, and iron oxide) in a rotary kiln at approximately 1450°C to produce an intermediate product called clinker. The clinker contains complex calcium silicate and aluminate compounds, tricalcium silicate (C_3S), dicalcium silicate (C_2S), tricalcium aluminate (C_3A), and tetracalcium aluminoferrite (C_4AF). The clinker is then ground with 3–5% gypsum ($CaSO_4 \cdot 2H_2O$), which is added as a retarder to control setting time. Without gypsum, cement would set almost instantaneously upon contact with water, making it unworkable. Hardening occurs through hydration reactions forming calcium silicate hydrate (C-S-H gel), the principal binding phase responsible for mechanical strength.

- Major types of cement and their distinguishing characteristics:
  - ✓ Ordinary Portland Cement (OPC): The most widely used cement globally, available in 33, 43, and 53 grades, denoting compressive strength in MPa at 28 days. Used in general construction, plastering, and concrete work.
  - ✓ Portland Pozzolana Cement (PPC): Produced by blending OPC clinker with pozzolanic materials such as fly ash, volcanic ash, or calcined clay (15–35%), along with gypsum. Generates less heat of hydration, making it suitable for mass concrete structures like dams. Also offers better sulphate resistance and economy.
  - ✓ Rapid Hardening Cement (RHC): Gains strength much faster than OPC, attaining in 3 days what OPC typically achieves in 28 days, due to finer grinding and higher C_3S content. Used in precast concrete, road repairs, and cold-weather concreting.
  - ✓ Low Heat Cement: Contains reduced C_3A and increased C_2S, producing less heat during hydration. Suitable for massive concrete structures such as gravity dams.
  - ✓ Sulphate Resistant Cement (SRC): Contains very low C_3A content (below 5%), making it resistant to sulphate attack from soil and groundwater. Used in marine structures, sewage plants, and foundations.
  - ✓ White Portland Cement: Manufactured using raw materials with minimal iron and manganese oxides, producing a white product. Used in architectural finishes, decorative concrete, and tile grouting.
  - ✓ Coloured Cement: Produced by adding mineral pigments to white or grey cement. Used in decorative flooring and ornamental construction.
  - ✓ High Alumina Cement (HAC): Manufactured using limestone and bauxite rather than clay, producing high alumina content. Resistant to high temperatures and chemical attack, used in refractory linings and industrial applications.
  - ✓ Expansive Cement: Contains additives causing controlled expansion during setting to offset shrinkage. Used in grouting, structural repairs, and prestressed concrete.
- Oil Well Cement: Specially designed to withstand high temperature and pressure conditions in oil and gas well drilling, with delayed setting characteristics.

Q 112. Ans. A

Explanation:

- **Statement 1 is correct** because in transverse waves, the displacement of particles occurs at right angles to the direction in which the wave travels. Light waves and waves on a stretched string are common examples.
- **Statement 2 is also correct** because in longitudinal waves, the particles oscillate back and forth in the same direction as the wave propagation. Sound waves in air are examples of longitudinal waves.
- **Statement 3 is incorrect** because sound waves in air propagate through compressions and rarefactions, making them longitudinal rather than transverse.

Q 113. Ans. A

Explanation:

- **Doppler Effect:** The idea of the expanding universe is based on the Doppler Effect observed in light coming from distant galaxies.
- When a source of light moves away from an observer, the wavelength of light increases and the spectral lines shift towards the red end of the spectrum. This phenomenon is known as **red shift**.
- According to the Doppler Effect for light, an increase in wavelength indicates recession of the source from the observer. Observations made by Edwin Hubble showed that most galaxies exhibit red shift, implying that they are moving away from Earth.
- Moreover, more distant galaxies show greater redshift, indicating faster recession velocities. This led to the conclusion that the universe is continuously expanding.
- **Stark Effect:** The Stark Effect refers to the splitting of spectral lines in the presence of an external electric field. It is related to atomic spectra and not to the expansion of the universe.
- **Zeeman Effect**: The splitting of spectral lines in the presence of a magnetic field. It helps in studying the magnetic properties of atoms and stars, but does not explain the expanding universe.

- **Raman Effect:** Raman Effect involves the scattering of light with a change in wavelength due to interaction with molecules. It is used in spectroscopy to study molecular structure and is unrelated to cosmic expansion.

Q 114. Ans. C

Explanation:

- **Pair 1 is correct:** For an object beyond the centre of curvature, the image is formed between F and C and is diminished.
- **Pair 2 is correct:** When the object is placed at the focus, reflected rays become parallel after reflection, and the image is formed at infinity.
- **Pair 3 is incorrect:** At the centre of curvature, the image formed is of the same size, not enlarged.
- **Pair 4 is correct:** When the object lies between the pole and focus, the image formed is virtual, erect and enlarged behind the mirror. Thus, three pairs are correctly matched.

Q 115. Ans. D

Explanation:

- **Assertion (A) is false:** Concave mirrors are not used as rear-view mirrors in vehicles because they may form enlarged and inverted images depending on the position of the object, which is unsuitable for safe driving.
- **Reason (R) is true:** Convex mirrors are preferred as rear-view mirrors because they always produce erect and diminished images. They also provide a wider field of view due to their outward curvature, enabling drivers to observe a larger area of traffic behind the vehicle. Thus, the assertion is false while the reason is true.

Q 116. Ans. A

Explanation:

- The refractive index of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in that medium. A higher refractive index means greater optical density and more bending of light. The refractive indices of the given substances are as follows:
  - ✓ Air: approximately 1.0003
  - ✓ Water: approximately 1.33
  - ✓ Crown glass: approximately 1.52
  - ✓ Diamond: approximately 2.42
- The refractive indices of some common substances are as follows:
  - ✓ Vacuum: 1.0000
  - ✓ Ice: approximately 1.31
  - ✓ Kerosene: approximately 1.44
  - ✓ Fused quartz: approximately 1.46
  - ✓ Turpentine oil: approximately 1.47
  - ✓ Rock salt: approximately 1.54
  - ✓ Carbon disulphide: approximately 1.63
  - ✓ Dense flint glass: approximately 1.65
  - ✓ Ruby: approximately 1.71
  - ✓ Sapphire: approximately 1.77

Q 117. Ans. A

Explanation:

- The upper portion must be a concave mirror because a concave mirror can form an enlarged image when the object is placed close to it. Therefore, the forehead appears magnified.
- The middle portion must be a plane mirror because a plane mirror always forms an image of the same size as the object. Hence, the middle part of the body appears unchanged.
- The lower portion must be a convex mirror because a convex mirror always forms a diminished and erect image. Therefore, the legs appear smaller.

Q 118. Ans. A

Explanation:

- A mirage is an optical illusion commonly observed on hot roads or deserts during summer afternoons.
- Air near the ground becomes hotter and rarer than the air above it. Thus, the refractive index of air decreases near the surface. Light rays travelling from distant objects pass through continuously varying layers of air and undergo successive refraction.
- At a certain stage, the angle of incidence becomes greater than the critical angle, and the rays suffer total internal reflection before reaching the observer's eyes.
- The reflected rays appear to come from the ground, creating an illusion of water on the road surface.

Q 119. Ans. B

Explanation:

- Optical fibres consist of a dense glass or plastic core surrounded by a comparatively rarer cladding.
- When light enters the core at an appropriate angle, it continuously undergoes total internal reflection at the interface between the core and cladding. As a result, light remains confined inside the fibre and travels long distances with minimal loss of energy.
- This principle forms the basis of high-speed optical communication systems, medical endoscopy and fibre-optic sensors.

Q 120. Ans. C

Explanation:

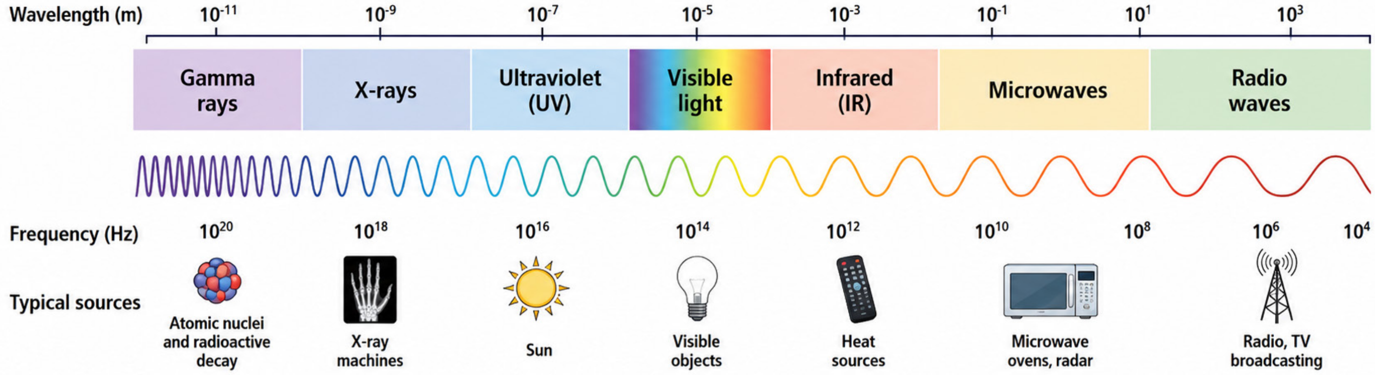
- At sunrise and sunset, sunlight travels a much longer distance through Earth's atmosphere before reaching the observer.
- The shorter wavelengths, such as blue and violet, are scattered away more strongly by atmospheric particles. Consequently, the longer wavelengths like red and orange dominate the light reaching our eyes. Therefore, the Sun appears reddish near the horizon.
- This phenomenon is explained by the scattering of light.

Q 121. Ans. A

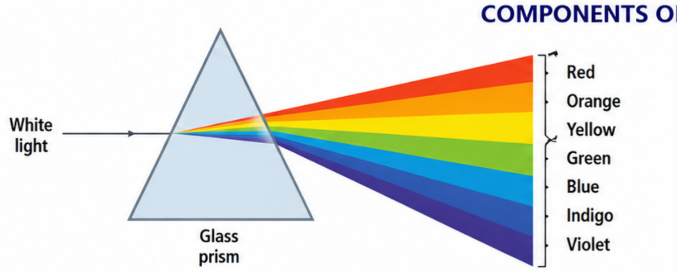
Explanation:

- Assertion (A) is true: The energy of light is inversely proportional to its wavelength. Since blue light has a shorter wavelength than red light, it possesses greater energy.
- Reason (R) is also true: In the visible spectrum, red light has a larger wavelength (about 700 nm), whereas blue light has a comparatively smaller wavelength (about 450 nm).

THE ELECTROMAGNETIC SPECTRUM



COMPONENTS OF WHITE LIGHT



Colour	Wavelength (nm)	Frequency (Hz)	Energy (eV)
Red	620 – 750	4.0×10^{14} – 4.8×10^{14}	1.65 – 2.00
Orange	590 – 620	4.8×10^{14} – 5.1×10^{14}	2.00 – 2.10
Yellow	570 – 590	5.1×10^{14} – 5.3×10^{14}	2.10 – 2.17
Green	495 – 570	5.3×10^{14} – 6.1×10^{14}	2.17 – 2.50
Blue	450 – 495	6.1×10^{14} – 6.7×10^{14}	2.50 – 2.75
Indigo	425 – 450	6.7×10^{14} – 7.1×10^{14}	2.75 – 2.92
Violet	380 – 425	7.1×10^{14} – 7.9×10^{14}	2.92 – 3.26

Q 122. Ans. B

Explanation:

- When light travels from air into water, it gets refracted. However, for rays attempting to emerge from water into air at angles greater than the critical angle, total internal reflection occurs.
- As a result, only rays within a certain angular range can enter the water and reach the swimmer's eyes. Therefore, objects above the surface are visible only through a circular region known as "Snell's window."
- Outside this region, the surface behaves like a mirror due to total internal reflection.

Q 123. Ans. B

Explanation:

- Myopia is a defect of vision in which a person can see nearby objects clearly, but distant objects appear blurred. This defect occurs because the image of distant objects is formed in front of the retina.
- In myopia (near-sightedness), the eye lens converges light rays too much, or the eyeball becomes elongated. As a result, the image of distant objects forms in front of the retina instead of on it.
- A concave lens is used to correct this defect. The concave lens diverges the incoming light rays slightly before they enter the eye. This enables the eye lens to focus the image correctly on the retina.
- Convex lenses are used to correct hypermetropia, not myopia. Cylindrical lenses are generally used for correcting astigmatism.

Q 124. Ans. D

Explanation:

- When red light and green light are combined in suitable proportions, they produce yellow light.

- The combination of green and blue light produces cyan. Cyan is therefore known as a secondary colour of light.
- When blue light combines with red light, the resultant colour is magenta.
- An equal combination of all three primary colours of light — red, green and blue — produces white light.

Additional Concept:

- The primary colours of light are: Red, Green and Blue
- The secondary colours formed by combining two primary colours are:
  - ✓ **Red + Green → Yellow**
  - ✓ **Green + Blue → Cyan**
  - ✓ **Blue + Red → Magenta**
- This process is known as additive colour mixing and is used in televisions, computer screens, mobile displays and digital imaging systems.

Q 125. Ans. C

Explanation:

- The soda-acid fire extinguisher operates on the principle of a rapid acid-carbonate reaction between sodium bicarbonate ($NaHCO_3$) solution and sulphuric acid (H_2SO_4), which are stored separately within the cylinder, typically the acid in a small inner glass vial and the bicarbonate solution in the outer chamber. When the extinguisher is inverted or struck, the vial breaks and the two reactants mix, producing the following reaction:
  - ✓ $2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O + 2CO_2\uparrow$
- The carbon dioxide (CO_2) generated is expelled under pressure through the nozzle as a stream directed at the fire. CO_2 extinguishes fire through two simultaneous mechanisms, it displaces and dilutes the oxygen surrounding the burning material, cutting off the oxygen supply essential for combustion, and being heavier than air, it forms a dense smothering blanket over the burning surface that prevents fresh oxygen from reaching the fuel. Additionally, CO_2 itself is non-combustible and chemically inert under fire conditions.
- Soda-acid extinguishers are suitable for Class A fires involving ordinary combustibles such as wood, paper, and cloth, but must not be used on electrical fires due to the water content, or on certain metal fires where CO_2 may react.

Q 126. Ans. A

Explanation:

- Common pigments used in phosphorescent materials include zinc sulphide (ZnS) and strontium aluminate. Use of zinc sulphide for safety-related products dates back to the 1930s. Copper-activated zinc sulphide shows intense and long-lasting phosphorescence, making it one of the most widely used phosphors historically.
- Phosphorescence is a form of photoluminescence in which a substance absorbs light energy, exciting electrons to higher energy states, and then re-emits light gradually over an extended period as electrons return to the ground state through a slow spin-forbidden transition, distinguishing it from fluorescence, which ceases almost immediately after the excitation source is removed.
- Phosphorescent materials (glow-in-the-dark, prolonged emission):
  - ✓ Zinc sulphide (ZnS), the most historically common phosphorescent material, used in X-ray screens, cathode ray tubes, paints, children's toys, and safety markings. Copper-doped zinc sulphide is also used in electroluminescent panels.
  - ✓ Strontium aluminate ($SrAl_2O_4$) doped with europium and dysprosium, the modern superior alternative to ZnS, approximately 10 times brighter and 10 times longer glowing. Used in exit signs, pathway markings, and safety signage.
  - ✓ Calcium sulphide (CaS) and barium sulphide (BaS), earlier phosphorescent materials now largely superseded.
- Fluorescent materials (immediate emission, ceases with excitation source):
  - ✓ Fluorescein, bright green fluorescent dye used in water tracing, medical diagnostics, and highlighter inks.
  - ✓ Rhodamine B, red fluorescent dye used in biological staining and laser applications.
  - ✓ Zinc orthosilicate (Zn_2SiO_4) doped with manganese, green fluorescent material used in early fluorescent lamps and cathode ray tubes.
  - ✓ Calcium halophosphate, white fluorescent material used in conventional fluorescent tube lamps.
- Bioluminescent materials (light produced by living organisms through chemical reactions):
  - ✓ Luciferin-luciferase system, responsible for

the glow of fireflies, where luciferin oxidation catalysed by luciferase produces light with near-100% efficiency.
  - ✓ Aequorin and GFP (Green Fluorescent Protein), found in jellyfish (Aequorea victoria) and widely used as biological research markers.
  - ✓ Bacterial luciferase, found in marine bacteria and bioluminescent fish.
- Chemiluminescent materials (light produced by chemical reactions without heat):
  - ✓ Luminol ($C_8H_7N_3O_2$), which produces blue light upon oxidation with hydrogen peroxide in alkaline conditions, used in forensic blood detection.
  - ✓ Oxalyl chloride and dye systems, forming the basis of glowsticks, where mixing chemicals produces sustained light.
- Thermoluminescent materials (light produced upon heating after prior irradiation):
  - ✓ Lithium fluoride (LiF) and calcium fluoride (CaF_2), used in radiation dosimeters.
  - ✓ Quartz and feldspar, used in thermoluminescence dating of archaeological samples.
- Radioluminescent materials (light produced by radioactive decay):
  - ✓ Tritium (^3H) with phosphor coatings, used in self-luminous watch dials, gun sights, and emergency signage.
  - ✓ Historically, Radium-226 with zinc sulphide, used in early luminous watch dials until abandoned due to radiation poisoning incidents such as the Radium Girls case.

Q 127. Ans. A

Explanation:

- Smart glass (also called electrochromic or switchable glass) changes its tint or transparency in response to an applied voltage through electrochromic materials, typically transition metal oxides like tungsten oxide (WO_3) or nickel oxide (NiO). When an electric voltage is applied, ions from a conductive coating move through the electrochromic layer, causing a chemical reaction that alters the oxidation state of the electrochromic material, thereby changing its optical properties. To return the glass to its clear state, the voltage is reversed, ions migrate back, the electrochromic material is oxidised, and transparency is restored.
- There is not just one smart glass technology, several distinct methods achieve transparency control through different scientific mechanisms. The primary technologies are Electrochromic (EC), Polymer Dispersed Liquid Crystal (PDLC), and Suspended Particle Device (SPD).
- Smart glass technologies:
  - ✓ Electrochromic (EC) glass: Uses transition metal oxides such as tungsten oxide or nickel oxide embedded in the glass structure. Applied voltage drives ion migration, changing oxidation state and optical behaviour. Used in architectural glazing, aircraft windows, and automobile sunroofs.
  - ✓ Polymer Dispersed Liquid Crystal (PDLC) glass: Consists of microscopic liquid crystal droplets suspended in a polymer matrix. When electric current is applied, the droplets align and permit light transmission, making the glass transparent. When current is removed, the droplets scatter light, making the glass opaque. Used in privacy partitions, conference rooms, and hospital screens.
  - ✓ Suspended Particle Device (SPD) glass: Uses microscopic particles that align in response to electrical signals, allowing rapid and variable control of transparency. Used in automotive sunroofs and skylights.
  - ✓ Thermochromic glass: Changes tint in response to temperature, not voltage. Materials such as vanadium dioxide (VO_2) undergo reversible phase transitions affecting transparency. Used in passive smart windows.
  - ✓ Photochromic glass: Darkens in response to ultraviolet light and clears when UV exposure stops. This is the same principle used in self-darkening spectacle lenses, involving silver halide (AgCl/AgBr) or organic photochromic compounds.
  - ✓ Related stimulus-responsive material phenomena:
  - ✓ Thermochromism: Colour or transparency change due to temperature variation. Example: mood rings and temperature-sensitive packaging indicators.
  - ✓ Photochromism: Reversible colour change due to light exposure. Example: photochromic lenses.
  - ✓ Halochromism: Colour change in response to pH. Example: litmus, phenolphthalein, methyl orange, and hydrangea pigments.
  - ✓ Solvatochromism: Colour change depending on solvent polarity, used in chemical analysis.

- ✓ Piezochromism: Colour change due to mechanical pressure, observed in certain transition metal complexes.

Q 128. Ans. B

Explanation:

- Cisplatin (cis-diamminedichloroplatinum(II)), with the chemical formula $[PtCl_2(NH_3)_2]$, is a square planar coordination compound with platinum (Pt) as the central metal atom in the +2 oxidation state, coordinated to two chloride ions (Cl^-) and two ammonia molecules (NH_3) in a cis configuration, meaning the two identical ligands occupy adjacent positions rather than opposite positions (the trans isomer, transplatin, is pharmacologically inactive).
- Cisplatin was first synthesised by Michele Peyrone in 1845 and was historically known as Peyrone's salt. Its anticancer properties were accidentally discovered by Barnett Rosenberg in 1965 during experiments on bacterial cell division, when platinum electrodes unexpectedly inhibited mitosis. It received US FDA approval in 1978 and remains a cornerstone chemotherapeutic drug for testicular, ovarian, bladder, lung, and head and neck cancers.

Q 129. Ans. D

Explanation:

- Ethinyl estradiol (EE) is a synthetic derivative of the naturally occurring female sex hormone oestradiol (estradiol), modified by the addition of an ethinyl group (–C≡CH) at the C-17 position, which prevents rapid hepatic degradation and makes it orally bioavailable. It is the most widely used synthetic oestrogen in combined oral contraceptive pills, typically formulated alongside a synthetic progestogen such as levonorgestrel, norethindrone, or desogestrel.
- Its contraceptive action operates through multiple complementary mechanisms, primarily by suppressing ovulation through negative feedback on the hypothalamic-pituitary axis, reducing secretion of FSH (follicle-stimulating hormone) and LH (luteinising hormone), thereby preventing follicular maturation and the LH surge required for ovulation. Additionally, it thickens cervical mucus, making sperm penetration difficult, and alters the endometrial lining, reducing the likelihood of implantation.
- Acetylsalicylic acid is aspirin, an analgesic, antipyretic, anti-inflammatory, and antiplatelet drug that inhibits cyclooxygenase (COX) enzymes.
- Sodium benzoate (C_6H_5COONa) is a food preservative used to inhibit microbial growth in acidic foods and beverages.
- Paracetamol (acetaminophen, $C_8H_9NO_2$) is an analgesic and antipyretic with no hormonal or contraceptive function.

Q 130. Ans. B

Explanation:

- **Pair 1 is incorrectly matched:** Aspirin (acetylsalicylic acid) is not an antacid. It is an analgesic, antipyretic, and anti-inflammatory drug belonging to the NSAID (Non-Steroidal Anti-Inflammatory Drug) class, functioning by irreversibly inhibiting COX-1 and COX-2 enzymes, thereby blocking prostaglandin synthesis. At low doses, it serves as an antiplatelet agent for cardiovascular protection. Antacids are entirely different substances, examples include sodium bicarbonate, magnesium hydroxide, aluminium hydroxide, and calcium carbonate, which neutralise excess gastric acid to relieve acidity and heartburn.
- **Pair 2 is correctly matched:** Paracetamol (acetaminophen, $C_8H_9NO_2$) is correctly classified as an antipyretic, and simultaneously an analgesic, acting on the hypothalamus to reduce fever and on central pain pathways to relieve mild to moderate pain. It is one of the most widely consumed over-the-counter medicines globally and is the preferred antipyretic for children due to its comparatively favourable safety profile relative to aspirin.
- **Pair 3 is incorrectly matched:** Penicillin is not a tranquilizer. It is a beta-lactam antibiotic, discovered by Alexander Fleming in 1928, that kills bacteria by inhibiting cell wall synthesis, specifically through irreversible binding to penicillin-binding proteins, preventing cross-linking of peptidoglycan chains in the bacterial wall and causing osmotic lysis. Tranquilizers are psychotherapeutic agents such as diazepam and chlorpromazine, entirely unrelated in pharmacological function.
- **Pair 4 is incorrectly matched:** Chloroquine is not an antiseptic. It is an antimalarial drug that acts by accumulating in the food vacuole of Plasmodium parasites and interfering with the detoxification of haem released during haemoglobin digestion, causing toxic haem accumulation that kills the parasite. It has also been used in rheumatoid arthritis and lupus for its immunomodulatory effects. Antiseptics are topically applied antimicrobial agents such as Dettol, iodine, and hydrogen

peroxide, whereas chloroquine is a systemically administered therapeutic drug with a completely different pharmacological classification.

Q 131. Ans. A

Explanation:

- Sodium benzoate (C_6H_5COONa) is the most widely used chemical preservative in packaged and processed food products, functioning by inhibiting the growth of bacteria, yeast, and moulds. It is effective primarily in acidic foods (pH below 4.5) because its active preservative form is benzoic acid (C_6H_5COOH), which penetrates microbial cell membranes in acidic conditions and interferes with the cell's ability to metabolise glucose and maintain pH homeostasis. Common applications include carbonated beverages, fruit juices, jams, pickles, sauces, and salad dressings. It is designated as E211 in the European food additive numbering system and is approved by FSSAI (Food Safety and Standards Authority of India) within specified limits.
- Sodium bicarbonate ($NaHCO_3$), baking soda, is used as a leavening agent in baking, producing CO_2 upon heating to make dough rise. It is also used as an antacid to neutralise stomach acidity but has no preservative function against microbial spoilage.
- Sodium citrate ($Na_3C_6H_5O_7$) is used as a buffering agent, emulsifier, and flavouring agent in processed foods and beverages, it controls pH and prevents syneresis in processed cheese. It is also used in oral rehydration solutions and as an anticoagulant in blood collection. It has no direct antimicrobial preservative role.
- Sodium acetate (CH_3COONa) is used primarily as a flavouring agent (providing a salt-and-vinegar taste) in snack foods like crisps and chips, and has limited antimicrobial activity as an acidity regulator. While acetic acid has some preservative properties in very acidic conditions (as in vinegar-based pickling), sodium acetate itself is not classified as a primary food preservative.

Q 132. Ans. C

Explanation:

- Aspartame ($C_{14}H_{18}N_2O_5$) is a dipeptide methyl ester composed of two amino acids, aspartic acid and phenylalanine, with approximately 200 times the sweetening power of sucrose. Upon digestion, aspartame is hydrolysed and releases phenylalanine as one of its breakdown products. This makes it entirely unsuitable for individuals suffering from phenylketonuria (PKU), a rare autosomal recessive metabolic disorder caused by deficiency of the enzyme phenylalanine hydroxylase, which normally converts phenylalanine into tyrosine. In PKU patients, phenylalanine accumulates to toxic levels in the blood and brain, causing severe intellectual disability, seizures, and neurological damage if untreated. Regulatory authorities worldwide require products containing aspartame to carry the warning "Contains Phenylalanine" specifically for this reason. Aspartame is marketed under brand names such as NutraSweet and Equal.
- Saccharin ($C_7H_5NO_3S$) was the first artificial sweetener, discovered in 1879 by Constantin Fahlberg, and is approximately 300–500 times sweeter than sucrose. It contains no phenylalanine and is safe for PKU patients. It is used in diet beverages, tabletop sweeteners, and pharmaceutical formulations. At high concentrations, it may produce a slightly bitter metallic aftertaste.
- Sucralose ($C_{12}H_{19}Cl_3O_8$) is produced by the selective chlorination of sucrose at three hydroxyl positions, making it approximately 600 times sweeter than sucrose. It is not metabolised significantly, passes through the digestive system largely unchanged, and contributes negligible calories. It contains no phenylalanine and is safe for PKU patients. It is marketed as Splenda.
- Sorbitol ($C_6H_{14}O_6$) is a sugar alcohol naturally present in fruits such as apples, pears, and prunes, and is approximately 60% as sweet as sucrose. It is partially metabolised and provides approximately 2.6 kcal/g, lower than sucrose. It is used in diabetic foods, chewing gum, and pharmaceutical syrups. It contains no phenylalanine and is safe for PKU patients, though excessive intake may cause osmotic diarrhoea.

Q 133. Ans. B

Explanation:

- Alum (potassium aluminium sulphate, $KAl(SO_4)_2 \cdot 12H_2O$) is the most widely used coagulant in water treatment and purification. When added to turbid water, alum dissolves and the aluminium ions (Al^{3+}) hydrolyse to form aluminium hydroxide $[Al(OH)_3]$, a gelatinous, sticky precipitate that acts as a coagulant by attracting and adsorbing suspended colloidal particles, clay, silt, bacteria, and organic matter onto its surface. These particles aggregate into larger masses called flocs, which become heavy enough to settle by gravity (sedimentation) or be removed through filtration. This process is known as coagulation-flocculation and is a fundamental step in municipal water treatment systems worldwide. Alum is also traditionally used at the household level in rural India for clarifying drinking water.

- Bleaching powder ($CaOCl_2$) is used in water treatment as a disinfectant and sterilising agent, releasing chlorine that kills bacteria and pathogens, but it does not function as a coagulant for suspended impurities.
- Sodium carbonate (Na_2CO_3), or washing soda, is used for water softening, precipitating calcium and magnesium ions responsible for hardness, but has no coagulation role.
- Potassium nitrate (KNO_3) has no water purification role. It is used mainly as a fertiliser, oxidising agent, and preservative, and excessive nitrate contamination in drinking water is itself hazardous due to the risk of methemoglobinaemia (blue baby syndrome).

Q 134. Ans. A

Explanation:

- TNT stands for 2,4,6-Trinitrotoluene ($C_7H_5N_3O_6$). It is produced by the nitration of toluene ($C_6H_5CH_3$) through sequential treatment with mixtures of concentrated nitric acid and sulphuric acid, introducing three nitro groups ($–NO_2$) at the 2, 4, and 6 positions of the aromatic ring. It was first synthesised by Julius Wilbrand in 1863, while its explosive properties were recognised by Carl Häussermann in 1891, after which it became the standard military explosive of the 20th century, particularly from World War I onward.
- TNT detonates through rapid exothermic decomposition, releasing enormous quantities of gases and energy in a very short time. Its explosive energy has become the standard benchmark for measuring explosive yield, giving rise to the unit "tonne of TNT equivalent", where 1 tonne of TNT releases approximately 4.184 gigajoules of energy.
- A major practical advantage of TNT is its relative insensitivity to shock and friction compared to more sensitive primary explosives such as nitroglycerin. It generally requires a detonator for initiation, making it significantly safer to handle, transport, and store.
- TNT is also commonly used in explosive mixtures, for example:
  - ✓ Composition B = RDX + TNT
  - ✓ Amatol = Ammonium nitrate + TNT

Q 135. Ans. B

Explanation:

- **Statement 1 is correct:** α-particles have very high ionising power because they are heavy charged particles consisting of two protons and two neutrons. A single α-particle can ionise thousands of gas atoms during its path.
- **Statement 2 is correct:** β-particles are much more penetrating than α-particles. While α-particles can be stopped by a thin aluminium sheet, β-particles can pass through a few millimetres of aluminium.
- **Statement 3 is incorrect:** γ-rays are electrically neutral electromagnetic waves and hence are not deflected by electric or magnetic fields. They travel at the speed of light.
- **Statement 4 is correct:** γ-rays possess the highest penetrating power among the three radiations and can pass through several centimetres of lead or iron sheets.

Q 136. Ans. A

Explanation:

- **Statement 1 is correct:** Cobalt-60, a radioactive source, emits gamma rays and is used in cancer treatment (radiotherapy).
- **Statement 2 is incorrect:** Carbon dating is based on the decay of carbon-14, not uranium to lead. Uranium-lead dating is used for geological timescales, like estimating the age of the Earth.
- **Statement 3 is correct:** Gamma radiation can be used to improve crop yield and prevent decay by irradiating seeds.
- **Statement 4 is incorrect:** It is gamma radiation (not β-particles) that is used industrially to detect flaws, due to its deep penetration ability.

Q 137. Ans. C

Explanation:

- Carbon-14 is a radioactive isotope of carbon that decays over time. By comparing the ratio of carbon-14 to carbon-12 in a fossil, scientists can estimate the time since the organism's death.
- Uranium-238 is used in geological dating (e.g., age of the Earth).
- Lead-206 is a decay product of uranium and is used in uranium-lead dating.
- Cobalt-60 is used in medical applications like radiotherapy.

Q 138. Ans. C

Explanation:

- The law of radioactive decay states that the number of decays per second (activity) is proportional to the number of radioactive atoms present at that moment.
- The half-life is the time it takes for half of the original atoms to decay, not all.
- The decay law is exponential, where λ is the decay constant.
  - ✓ $N(t) = N_0 \exp(-\lambda t)$.
- The decay rate depends on the number of atoms, not just the elapsed time.

Q 139. Ans. B

Explanation:

- Uranium-235 is a fissile isotope that can sustain a chain reaction. When a neutron strikes a U-235 nucleus, it undergoes fission, releasing energy and more neutrons that propagate the reaction.
- U-234, U-238, and U-236 are not as readily fissile as U-235, which is critical for sustaining a chain reaction in a reactor.

Q 140. Ans. B

Explanation:

- **Statement 1 is incorrect:** The Hydrogen Bomb relies on fusion for its primary energy release, though a small fission bomb triggers it.
- **Statement 2 is correct:** Fusion releases far more energy per unit mass than fission, which is why Hydrogen Bombs are so powerful.
- **Statement 3 is incorrect:** Fusion requires extremely high temperatures and pressures, much higher than those needed for fission, making it a thermonuclear process.

Q 141. Ans. B

Explanation:

- Acid rain is formed when sulphur dioxide (SO_2) and nitrogen oxides (NO_x, primarily NO and NO_2) are released into the atmosphere and undergo chemical reactions with water, oxygen, and atmospheric oxidising agents to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3). These acids dissolve in atmospheric moisture and fall as precipitation with a pH significantly below the normal 5.6 of unpolluted rainwater.
- The principal reactions are:
  - ✓ $SO_2 + H_2O \rightarrow H_2SO_3$ (sulphurous acid, further oxidised to H_2SO_4)
  - ✓ $2SO_2 + O_2 \rightarrow 2SO_3$
  - ✓ $SO_3 + H_2O \rightarrow H_2SO_4$ (sulphuric acid)
  - ✓ $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$ (nitric acid)
- SO_2 is primarily emitted by coal-fired thermal power plants, metal smelters, petroleum refining, and volcanic eruptions, while NO_x originates mainly from vehicular exhaust, industrial combustion, and thermal power generation.
- The ecological and structural consequences of acid rain are extensive:
  - ✓ Acidification of lakes and rivers, killing fish and other aquatic organisms
  - ✓ Leaching of essential soil nutrients such as calcium and magnesium
  - ✓ Damage to forests, particularly coniferous ecosystems
  - ✓ Corrosion of buildings and monuments, especially those made of marble and limestone
  - ✓ A significant example is the Taj Mahal, where concern over atmospheric SO_2 emissions from nearby industrial activity led to anti-pollution measures in the Taj Trapezium Zone.
- Carbon monoxide (CO) and methane (CH_4) are pollutants but do not produce acid rain.
- Hydrogen and oxygen combine to form water, not acids.
- Chlorofluorocarbons (CFCs) are associated with stratospheric ozone depletion, a separate environmental issue.

Q 142. Ans. B

Explanation:

- Chlorofluorocarbons (CFCs), historically used as refrigerants, aerosol propellants, foam-blowing agents, and cleaning solvents, are chemically stable in the troposphere, which allows them to persist long enough to migrate into the stratosphere. There, high-energy ultraviolet (UV-C) radiation photolytically cleaves the carbon-chlorine bond, releasing highly reactive chlorine radicals (Cl•):
  - ✓ $CF_2Cl_2 + UV \rightarrow CF_2Cl\bullet + Cl\bullet$
- The chlorine radical then destroys ozone through a catalytic chain reaction:
  - ✓ $Cl\bullet + O_3 \rightarrow ClO\bullet + O_2$
  - ✓ $ClO\bullet + O\bullet \rightarrow Cl\bullet + O_2$
  - ✓ Net reaction: $O_3 + O\bullet \rightarrow 2O_2$
- The crucial feature is that the chlorine radical is regenerated at the end of the cycle rather than consumed, making it a true catalyst. Because of this catalytic regeneration, a single chlorine atom can

destroy approximately 100,000 ozone molecules before being permanently trapped in stable reservoir compounds such as hydrogen chloride (HCl) or chlorine nitrate ($ClONO_2$).

- This catalytic amplification explains why even relatively small atmospheric concentrations of CFCs can produce severe stratospheric ozone depletion, particularly over polar regions, leading to the ozone hole phenomenon. Ozone depletion increases the penetration of harmful ultraviolet-B (UV-B) radiation to Earth's surface, raising risks of skin cancer, cataracts, immune suppression, crop damage, and marine ecosystem disruption.
- The Montreal Protocol (1987), regarded as one of the most successful international environmental treaties, mandated the global phaseout of CFCs and other ozone-depleting substances, leading to gradual recovery of the ozone layer.

Q 143. Ans. B

Explanation:

- Magnesium hydroxide ($Mg(OH)_2$), commonly known as Milk of Magnesia, is a widely used antacid that neutralises excess hydrochloric acid (HCl) in the stomach through the reaction:
  - ✓ $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$
- It raises gastric pH and provides relief from acidity, heartburn, and indigestion. Being a relatively mild and slow-acting base, it produces less rebound acidity than stronger rapidly acting antacids. At higher doses, it also functions as a laxative by drawing water into the intestine through osmotic action.
- Other commonly used antacids include:
  - ✓ Aluminium hydroxide [$Al(OH)_3$]
  - ✓ Calcium carbonate ($CaCO_3$)
  - ✓ Sodium bicarbonate ($NaHCO_3$), which acts rapidly but is unsuitable for prolonged use due to sodium loading and risk of metabolic alkalosis
- Many commercial preparations combine magnesium hydroxide and aluminium hydroxide to balance magnesium's laxative effect with aluminium's constipating tendency.
- Sodium thiosulphate ($Na_2S_2O_3$) is used as a photographic fixing agent, cyanide antidote, and analytical reagent, not as an antacid.
- Calcium carbide (CaC_2) reacts violently with water to produce acetylene gas, making it hazardous for ingestion.
- Ammonium chloride (NH_4Cl) is an acidic salt and would worsen gastric acidity rather than neutralise it.

Q 144. Ans. C

Explanation:

- **Option 1 is incorrect:** Colloidal particles are not dissolved at the molecular level, that description applies to true solutions. Colloidal particles consist of aggregates of many molecules or large macromolecules and remain dispersed due to electrostatic repulsion between similarly charged particles and the surrounding solvation layer, not because of true molecular dissolution.
- **Option 2 is incorrect:** Colloidal particles cannot be separated by ordinary filter paper, because their particle size (1–1000 nm) is smaller than the pore size of normal filter paper. They can, however, be separated using dialysis membranes, ultrafiltration, or ultracentrifugation.
- **Option 3 is correct:** Colloidal systems are heterogeneous mixtures in which dispersed particles have sizes ranging from 1 to 1000 nanometres (nm), intermediate between true solutions (particle size below 1 nm) and suspensions (particle size above 1000 nm). This intermediate particle size is responsible for the Tyndall effect, the scattering of a beam of light by colloidal particles, making the light path visible when observed from the side. The phenomenon occurs because colloidal particles are of a size comparable to the wavelength of visible light, causing effective light scattering. Classic examples include the visible beam of sunlight through dust particles in air, the bluish tinge of dilute milk, and the scattering of headlight beams in fog.
- **Option 4 is incorrect:** Colloidal systems are not restricted to solid-in-liquid dispersions. They encompass multiple categories:
  - ✓ Aerosols: solid or liquid dispersed in gas (smoke, fog, mist)
  - ✓ Foams: gas dispersed in liquid or solid (soap foam, pumice stone)
  - ✓ Emulsions: liquid dispersed in liquid (milk, mayonnaise)
  - ✓ Gels: liquid dispersed in solid (gelatin, cheese)
  - ✓ Sols: solid dispersed in liquid (paint, starch solution)

Q 145. Ans. C

Explanation:

- Cyanide leaching (MacArthur-Forrest process) is the principal industrial method for extracting gold from its ores, particularly low-grade ores where gold occurs in finely disseminated form. The process involves treating crushed gold ore with a dilute solution of sodium cyanide (NaCN) in the presence of oxygen and water, which selectively dissolves gold through the reaction:

- ✓ $4Au + 8NaCN + O_2 + 2H_2O \longrightarrow 4Na[Au(CN)_2] + 4NaOH$

- Gold forms a stable, soluble sodium dicyanoaurate(I) complex $[Na[Au(CN)_2]]$, while the gangue remains undissolved.
- The dissolved gold is then recovered by methods such as:
  - ✓ Cementation with zinc dust (Merrill-Crowe process), where zinc displaces gold from the cyanide complex
  - ✓ Adsorption onto activated carbon (carbon-in-pulp / carbon-in-leach processes), followed by recovery
- The recovered metal is subsequently refined, often electrolytically, to obtain high-purity gold.
- Developed in 1887, this process remains the dominant industrial technology for gold extraction.
- Other metals use different extraction routes:
  - ✓ Copper: Smelting of copper pyrites ($CuFeS_2$) or solvent extraction-electrowinning (SX-EW) for oxide ores
  - ✓ Iron: Blast furnace reduction of haematite (Fe_2O_3) or magnetite (Fe_3O_4) using coke
  - ✓ Zinc: Roasting of zinc sulphide (ZnS) to ZnO, followed by reduction or electrolytic extraction

Q 146. Ans. B

Explanation:

- Graphene is a single layer of carbon atoms tightly bound in a two-dimensional hexagonal honeycomb lattice. It is the basic structural element of other allotropes, including graphite, charcoal, carbon nanotubes, and fullerenes.
- Unlike diamond, where carbon is sp^3 hybridised to form a rigid 3D insulator, each carbon atom in graphene is sp^2 hybridised and bonded to three other carbon atoms. The fourth valence electron is delocalised, which allows graphene to be an exceptional conductor of electricity and heat. Despite being only one atom thick, it is incredibly strong, lightweight, and flexible, making it a revolutionary material in nanotechnology and electronics.

Q 147. Ans. C

Explanation:

- A hydrogen fuel cell generates electricity by combining hydrogen gas (fuel) and oxygen gas (oxidiser) electrochemically.
- At the anode, hydrogen molecules are split into protons and electrons. The electrons flow through an external circuit, creating an electric current, while the protons pass through a proton exchange membrane to the cathode. At the cathode, protons, electrons, and oxygen combine to form pure water (H_2O) as the sole harmless by-product.
- Because they produce zero greenhouse gas emissions during operation, fuel cells are considered a critical technology for green energy and sustainable transportation.

Q 148. Ans. B

Explanation:

- Iodine solution is the standard chemical reagent used to test for the presence of starch in various food items. When a few drops of iodine tincture are added to a milk sample adulterated with starch, the solution turns an intense blue-black colour due to the formation of a starch-iodine complex.
- Pure milk, which naturally lacks starch, will not undergo this colour change and will merely take on the yellowish-brown tint of the iodine solution itself.
- Benedict's solution and Fehling's solution are used to test for reducing sugars (like glucose), while nitric acid is traditionally used in the xanthoproteic test to detect proteins.

Q 149. Ans. D

Explanation:

- The Meissner effect is a defining and strictly quantum mechanical characteristic of superconductivity. When a material is cooled below its critical temperature and transitions into a superconducting state, it does two things: it loses all electrical resistance, and it exhibits perfect diamagnetism.
- Perfect diamagnetism means the superconductor actively excludes all external magnetic fields from its interior. If a magnetic field is applied, the superconductor generates persistent surface electrical currents. These currents produce a magnetic field exactly equal and opposite to the applied field, effectively cancelling it out inside the material and expelling the magnetic flux lines.

- This complete expulsion of magnetic fields is what allows a magnet to levitate perfectly stably above a superconductor (quantum levitation), which is the foundational physics principle behind modern frictionless maglev trains and the powerful superconducting electromagnets used in MRI machines. This magnetic expulsion proves that a superconductor is not just a classical "perfect conductor," but a completely distinct thermodynamic state of matter.

Q 150. Ans. A

Explanation:

- **Pair 1 is correctly matched:** Electroplating involves depositing a thin coating of one metal over another using electric current to improve appearance, corrosion resistance or durability.
- **Pair 2 is correctly matched:** Electrolysis is the process in which electric current brings about a non-spontaneous chemical reaction, such as the decomposition of water into hydrogen and oxygen.
- **Pair 3 is correctly matched:** A fuel cell converts the chemical energy of a fuel directly into electrical energy through electrochemical reactions without combustion.
- **Pair 4 is incorrectly matched:** Galvanisation is the process of coating iron or steel with zinc to prevent rusting. Purification of copper by electrolysis is known as electrorefining.